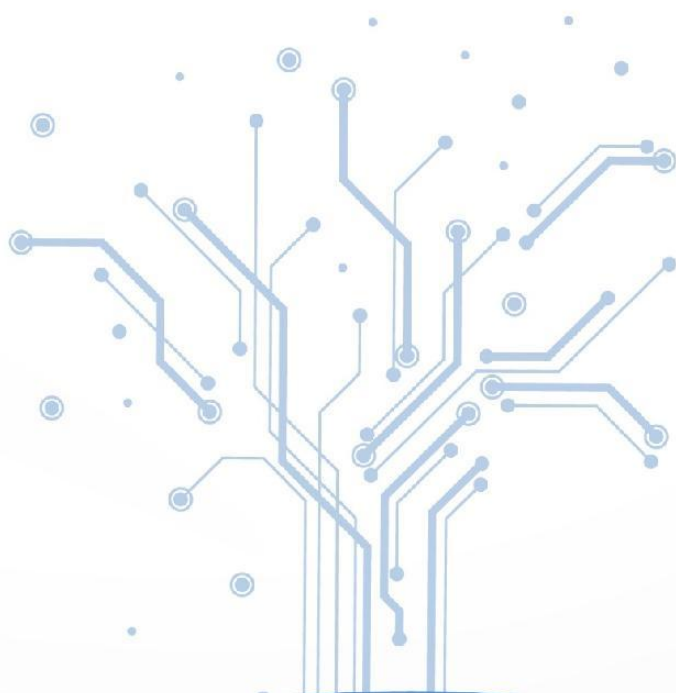




User Manual of the OpenAgri Web Platform

Functional descriptions for each illustrated platform page and service



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Table of Contents

1. Platform overview	6
2. OpenAgri Dashboard	7
2.1. Dashboard landing page	7
2.1.1. Purpose	7
2.1.2. Functional description	7
2.1.3. User actions	7
2.2. Farm Calendar	8
2.2.1. Purpose	8
2.2.2. Functional description	8
2.2.3. User actions	8
2.3. Register new activity – Activity type selection	9
2.3.1. Purpose	9
2.3.2. Functional description	9
2.3.3. User actions	9
2.4. Activity form	10
2.4.1. Purpose	10
2.4.2. Functional description	10
2.4.3. User actions	10
2.5. Farms	11
2.5.1. Purpose	11
2.5.2. Functional description	11
2.5.3. User actions	11
2.6. Farm parcels	12
2.6.1. Purpose	12
2.6.2. Functional description	12
2.6.3. User actions	12
2.7. Farm Animals	13
2.7.1. Purpose	13
2.7.2. Functional description	13
2.7.3. User actions	13
2.8. Agricultural machines	14
2.8.1. Purpose	14
2.8.2. Functional description	14

2.8.3.	User actions	14
2.9.	Crops.....	15
2.9.1.	Purpose.....	15
2.9.2.	Functional description	15
2.10.	Fertilizers	16
2.10.1	Purpose.....	16
2.10.2	Functional description	16
2.11.	Pesticides	17
2.11.1	Purpose.....	17
2.11.2	Functional description	17
3.	Reporting Service.....	18
3.1.	Compost operations	18
3.1.1.	Purpose.....	18
3.1.2.	Functional description	18
3.1.3.	User actions	18
3.2.	Farm animals report	19
3.2.1.	Purpose.....	19
3.2.2.	Functional description	19
3.2.3.	User actions	19
3.3.	Irrigation operations.....	20
3.3.1.	Purpose.....	21
3.3.2.	Functional description	21
3.3.3.	User actions	21
3.4.	Field notebook.....	21
3.4.1.	Purpose.....	21
3.4.2.	Functional description	21
4.	Irrigation Management	23
4.1.	Eto Calculator	23
4.1.1.	Purpose.....	23
4.1.2.	Functional description	23
4.1.3.	How to Interpret the Chart.....	24
4.1.4.	Operational Use.....	24
4.1.5.	User actions	24
4.2.	Upload Dataset	25
4.2.1.	Purpose.....	25

4.2.2.	Functional description	25
4.2.3.	User actions	25
4.3.	Soil Moisture Analysis.....	26
4.3.1.	Purpose.....	26
4.3.2.	Functional description	26
4.3.3.	Detected Events.....	26
4.3.4.	Reading High-Dose Irrigation Events	27
4.3.5.	Reading Saturation Days.....	27
4.3.6.	User actions	27
4.4.	Soil types.....	28
4.4.1.	Purpose.....	28
4.4.2.	Functional description	28
5.	Pest and disease management.....	29
5.1.	Pests.....	29
5.1.1.	Purpose.....	29
5.1.2.	Functional description	29
5.1.3.	User actions	29
5.2.	Growing degree days.....	30
5.2.1.	Purpose.....	30
5.2.2.	Functional description	30
5.2.3.	Workflow in the Platform.....	31
5.2.4.	Decision Support Value	31
5.2.5.	Practical Interpretation	31
5.2.6.	Purpose.....	32
5.2.7.	Functional description	32
5.2.8.	User actions	32
5.3.	Fuzzy Risk Engine	32
5.3.1.	Purpose.....	33
5.3.2.	How the Engine Works	33
5.3.3.	User actions	34
5.4.	Risk forecast.....	34
5.4.1.	Purpose.....	34
5.4.2.	Functional description	34
5.4.3.	User actions	34
6.	Weather data.....	35

6.1.1.	Purpose	35
6.1.2.	Functional description	35
6.1.3.	User actions	35

List of figures

Figure 1.	Dashboard landing page	7
Figure 2.	Monthly calendar view.	8
Figure 3.	Register new activity - activity type selection.	9
Figure 4.	Register new activity - activity form.	10
Figure 5.	Farm's page	11
Figure 6.	Farm parcels page.....	12
Figure 7.	Farm animals page.....	13
Figure 8.	Agricultural machines page	14
Figure 9.	Crop types form.....	15
Figure 10.	Fertilizers page.....	16
Figure 11.	Pesticides page	17
Figure 12.	Compost operations page	18
Figure 13.	Farm animals report..	19
Figure 14.	Irrigation operations page	20
Figure 15.	Irrigation operations report.....	20
Figure 16.	Field notebook page	21
Figure 17.	Field notebook.....	22
Figure 18.	ETo Calculator.....	23
Figure 19.	Upload Dataset	25
Figure 20.	Soil Moisture Analysis.....	26
Figure 21.	Soil Types form	28
Figure 22.	Pests page.....	29
Figure 23.	Growing Degree Days	30
Figure 24.	Treat model definition.....	31
Figure 25.	Fuzzy rules	32
Figure 26.	Threat models.....	33
Figure 27.	Risk forecast page.....	34
Figure 28.	Weather data table.....	35

1. Platform overview

The OpenAgri web platform is an integrated graphical interface that brings together the reusable OpenAgri services into a single Agricultural Digital Solution. It is intended for end users such as farmers, agronomists, advisors, farm managers, and pilot teams who need to record farming activities, manage farm assets, generate reports, analyse irrigation, monitor pest and disease risk, and use weather-based decision support.

The platform follows a modular microservice approach. The Dashboard provides the user interface, while the domain-specific services provide reusable backend functionality through APIs. The GateKeeper component acts as the secure access and routing point between the frontend and backend services. Several services exchange data using OpenAgri Common Semantic Model (OCSM) concepts and JSON-LD representations, which supports interoperability and reuse of records across modules.

Service	Main purpose	Illustrated pages	Key outputs
OpenAgri User Dashboard	Integrated user interface for all services, responsive/PWA, access through GateKeeper.	Dashboard landing page	Single point of access to Farm Calendar, Irrigation, Pest and Disease, Weather and Reporting modules.
Digital Farm Calendar	Digital recording of farm activities, observations, assets, farms, parcels, animals, crops, machinery and materials.	Farm calendar, activity registration, farms, farm parcels	Structured farm records, activity history, OCSM/JSON-LD data and API access.
Reporting Service	Automatic extraction of reports from recorded farm data.	Compost operations, farm animals, irrigation operations	Compost, livestock, irrigation and other domain-specific reports.
Irrigation Management	Decision-support calculators and analytics for irrigation planning.	ETo Calculator, Upload Dataset, Soil Moisture Analysis	ETo chart, soil moisture analysis, irrigation events, high-dose irrigation days, field capacity and stress indicators.
Pest and Disease Management	Rule-based and model-based pest/disease decision support.	Pests, Growing Degree Days, Threat Models, Risk Forecast	Pest definitions, GDD values, reusable threat models and risk forecast charts.
Weather Data Service	Forecast, historical weather data and agriculture-specific weather indicators.	Weather Data	5-day forecasts, local cache, THI, spraying suitability and UAV flight suitability.
OpenAgri User Dashboard	Integrated user interface for all services, responsive/PWA, access through GateKeeper.	Dashboard landing page	Single point of access to Farm Calendar, Irrigation, Pest and Disease, Weather and Reporting modules.
Digital Farm Calendar	Digital recording of farm activities, observations, assets, farms, parcels, animals, crops, machinery and materials.	Farm calendar, activity registration, farms, farm parcels	Structured farm records, activity history, OCSM/JSON-LD data and API access.

2. OpenAgri Dashboard

2.1. Dashboard landing page

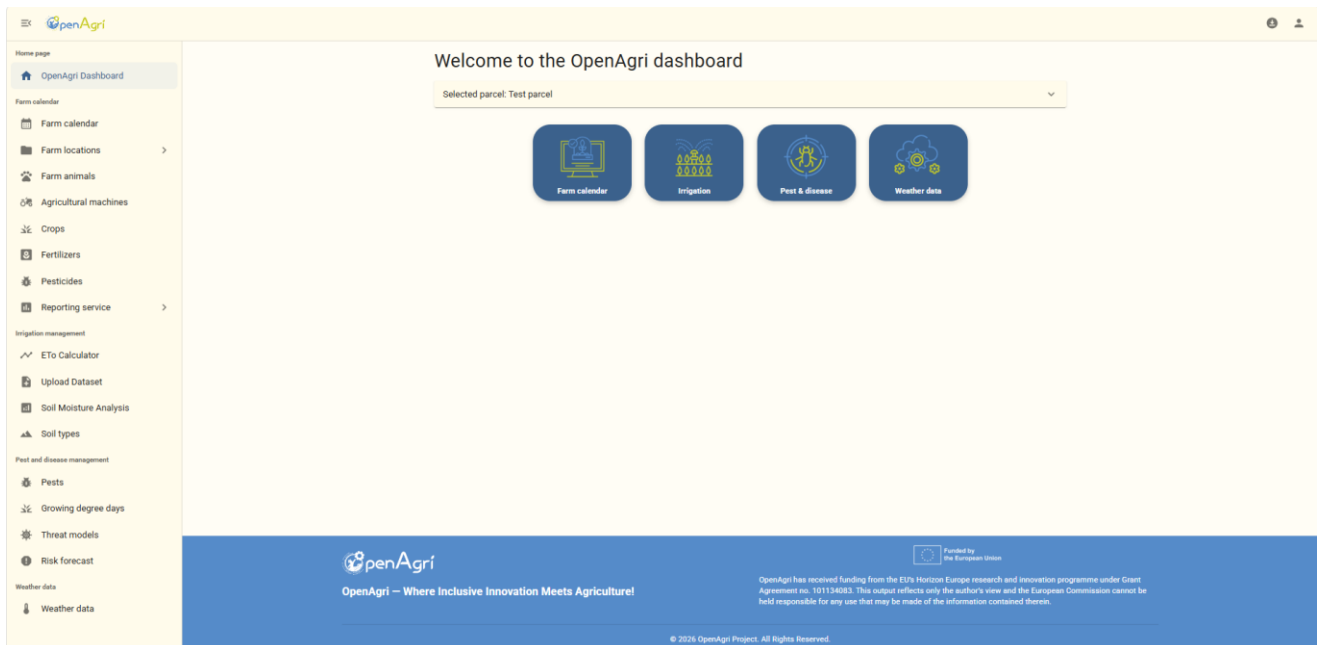


Figure 1. Dashboard landing page

2.1.1. Purpose

Provides the main entry point after login and gives the user quick access to the core OpenAgri modules.

2.1.2. Functional description

The landing page presents the selected farm parcel context at the top of the screen and exposes shortcuts to Farm Calendar, Irrigation, Pest and Disease Management, and Weather Data.

The left navigation menu groups the web platform into functional areas: home page, farm calendar, farm locations, reporting service, irrigation management, pest and disease management, and weather data.

The page represents the integrated OpenAgri Dashboard service. It combines the reusable OpenAgri services into a single graphical interface for farmers, agronomists, advisors, and farm managers. The dashboard uses a microservice architecture in which the backend services remain modular, while access is coordinated through the GateKeeper component.

2.1.3. User actions

- Select or change the active parcel from the parcel selector.
- Open a module directly from the large shortcut buttons.
- Navigate to service-specific pages using the left menu.

2.2. Farm Calendar

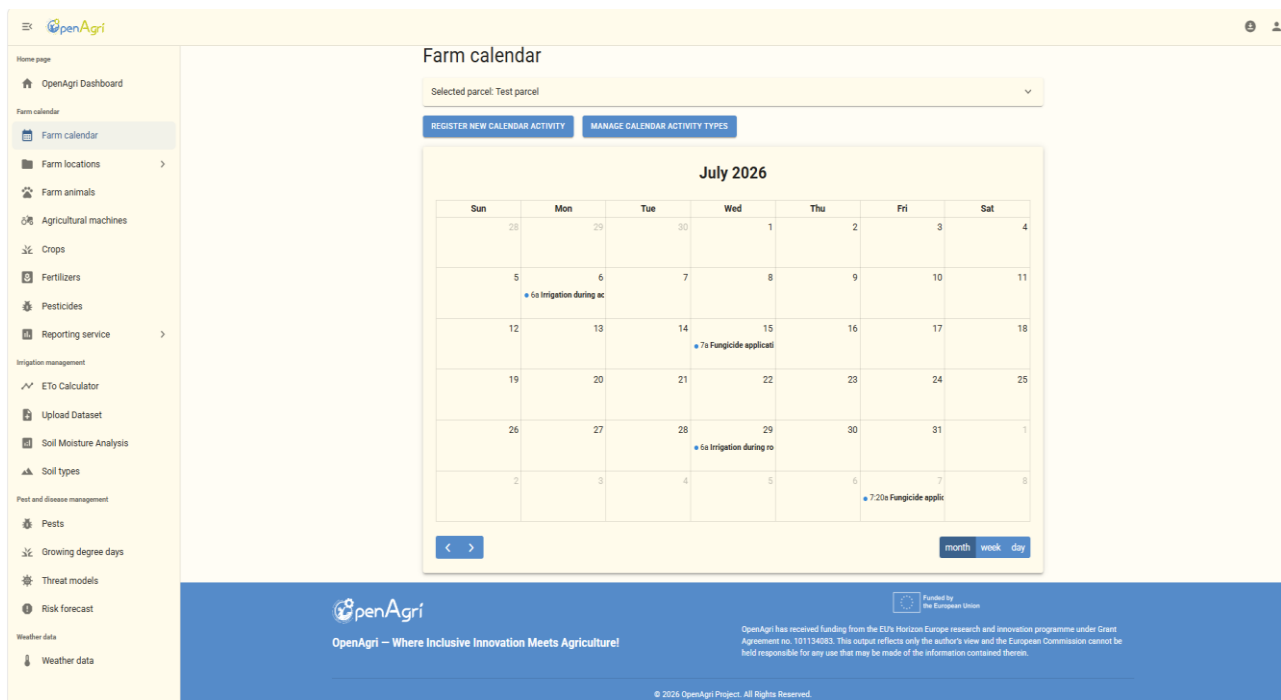


Figure 2. Monthly calendar view.

2.2.1. Purpose

Displays farm operations in a monthly calendar and supports planning, registration, and review of activities.

2.2.2. Functional description

The Farm Calendar service digitises farm records by allowing users to log day-to-day activities, observations, alerts, field operations, parcel activities, and asset-related operations.

Each registered operation appears on the calendar and can be colour-coded by activity type. The calendar view is linked to farm assets and parcels, so the same record can be reused by reporting, irrigation, and pest/disease workflows.

The service also exposes a REST API and interoperable JSON-LD/OCSM data models, enabling other OpenAgri services or external applications to reuse the same farm activity records.

2.2.3. User actions

- Select a parcel to filter activities for the relevant location.
- Use the calendar controls to change month, week, or day view.
- Register a new calendar activity or manage the set of calendar activity types

2.3. Register new activity – Activity type selection

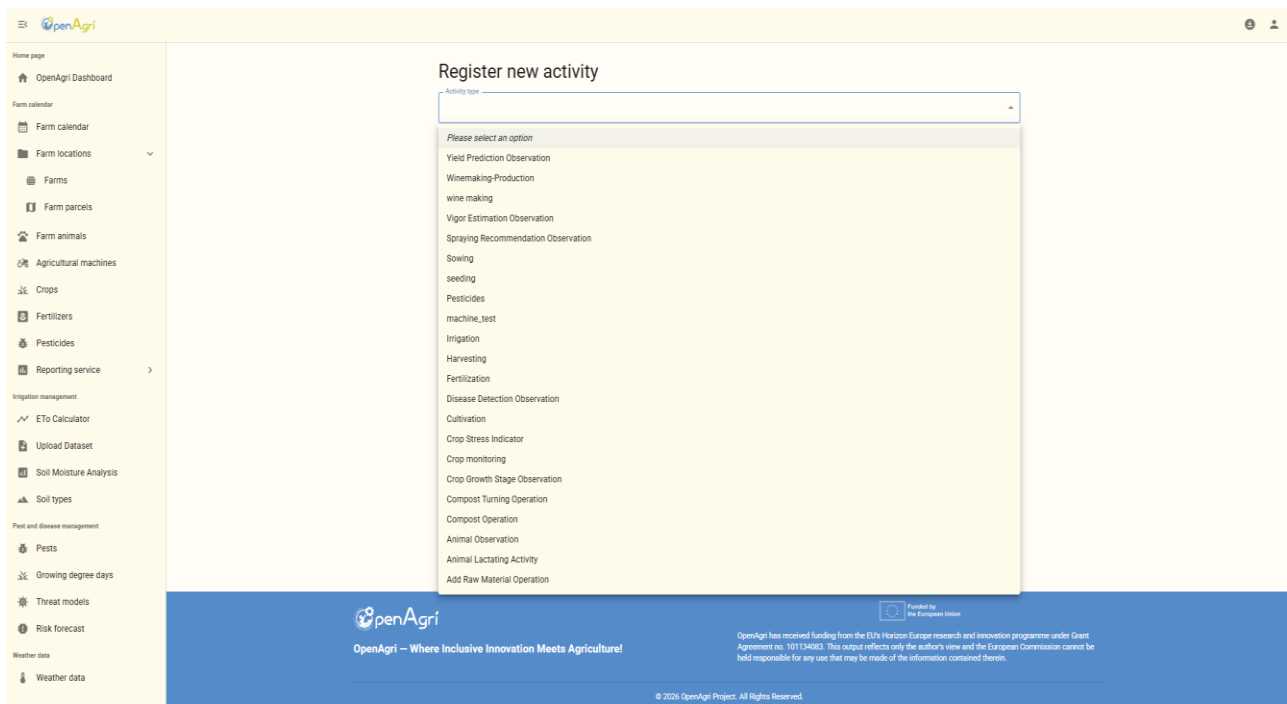


Figure 3. Register new activity - activity type selection.

2.3.1. Purpose

Allows the user to choose the type of farm activity or observation that will be registered.

2.3.2. Functional description

The activity type selector defines which form and metadata fields will be shown to the user. The list includes agronomic operations, observations, irrigation, pesticide and fertilisation operations, sensor-related records, compost operations, animal activities, and custom activities.

This approach supports a flexible but structured farm log. Standard activity types promote consistent recording and reporting, while custom options allow local practices and pilot-specific workflows to be captured without changing the whole platform.

After the type is selected, the platform opens a form with fields relevant to that activity. The resulting record becomes part of the farm calendar and can later be used by reporting and analytics services.

2.3.3. User actions

- Open Register new calendar activity from the Farm Calendar page.
- Choose the relevant activity or observation type.
- Continue to the generated form and complete the required information.

2.4. Activity form

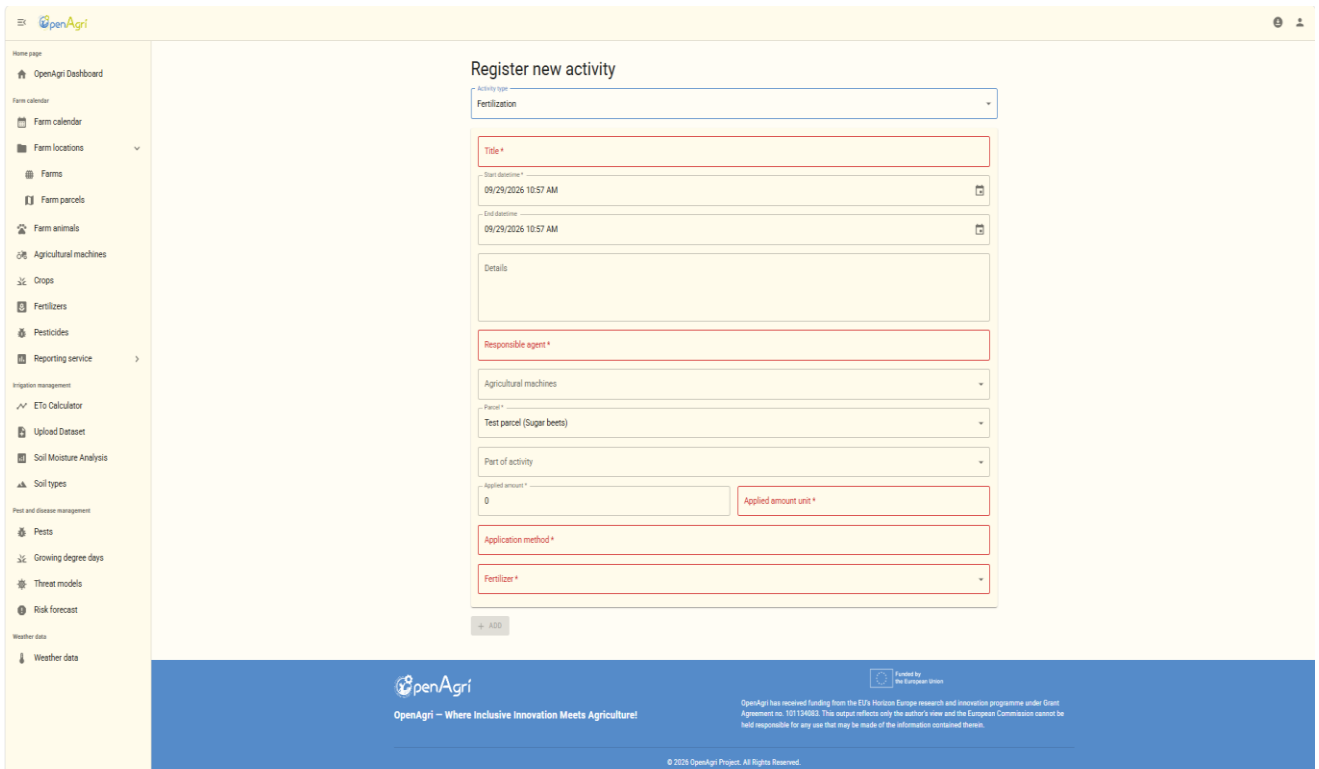


Figure 4. Register new activity - activity form.

2.4.1. Purpose

Captures the detailed metadata required to create a farm activity record.

2.4.2. Functional description

The activity form collects common activity fields such as title, start and end date, details, responsible agent, agricultural machine, parcel, and phase of activity. Mandatory fields are highlighted to guide complete data entry.

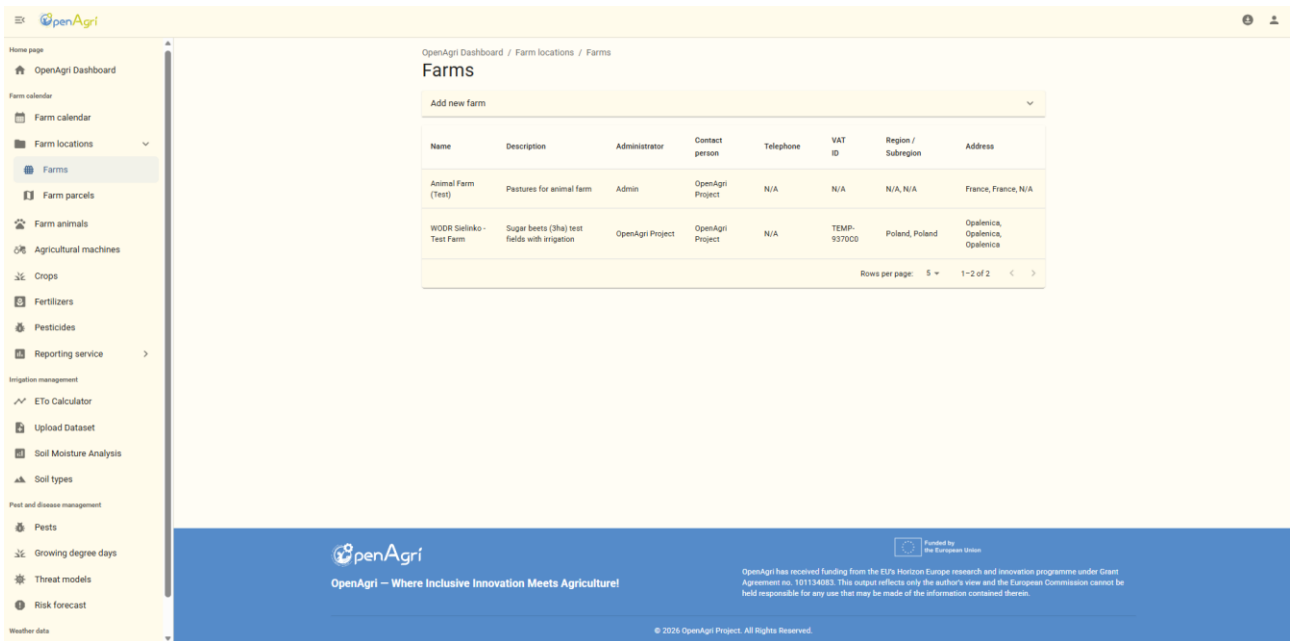
The form supports the OpenAgri principle of collecting operational data once and reusing it many times. A completed activity can appear in the calendar, feed reporting workflows, and provide contextual information for decision-support services.

Records created through the Farm Calendar are stored in a database and are designed to be represented according to the OpenAgri Common Semantic Model, which enables interoperability between modules.

2.4.3. User actions

- Fill in the required title and time period.
- Describe the operation or observation in the details field.
- Associate the record with a responsible agent, machine, parcel, and activity phase when applicable.
- Submit the form to add the record to the calendar.

2.5. Farms



OpenAgri Dashboard / Farm locations / Farms

Farms

Add new farm

Name	Description	Administrator	Contact person	Telephone	VAT ID	Region / Subregion	Address
Animal Farm (Test)	Pastures for animal farm	Admin	OpenAgri Project	N/A	N/A	N/A, N/A	France, France, N/A
WOGR Seliwko - Test Farm	Sugar beets (3ha) test fields with irrigation	OpenAgri Project	OpenAgri Project	N/A	TEMP-937000	Poland, Poland	Opalenica, Opalenica, Opalenica

Rows per page: 5 1-2 of 2

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Figure 5. Farm's page

2.5.1. Purpose

Maintains the registry of farms that are available to the user in the platform.

2.5.2. Functional description

The Farms page is part of the Farm Calendar and farm location management functionality. It provides a structured table of farm records, including name, description, address information, contact person, telephone, VAT ID, regional information, and additional notes.

Farm records provide the organisational context used by parcels, operations, reports, and service calculations. A well-maintained farm registry improves traceability and ensures that reports and analytics are linked to the correct farm entity.

The page also supports adding new farms, which extends the set of locations and assets available for operational recording.

2.5.3. User actions

- Review existing farms and their administrative/contact information.
- Add a new farm from the page control.
- Use farm records as the parent context for parcels and activities.

2.6. Farm parcels

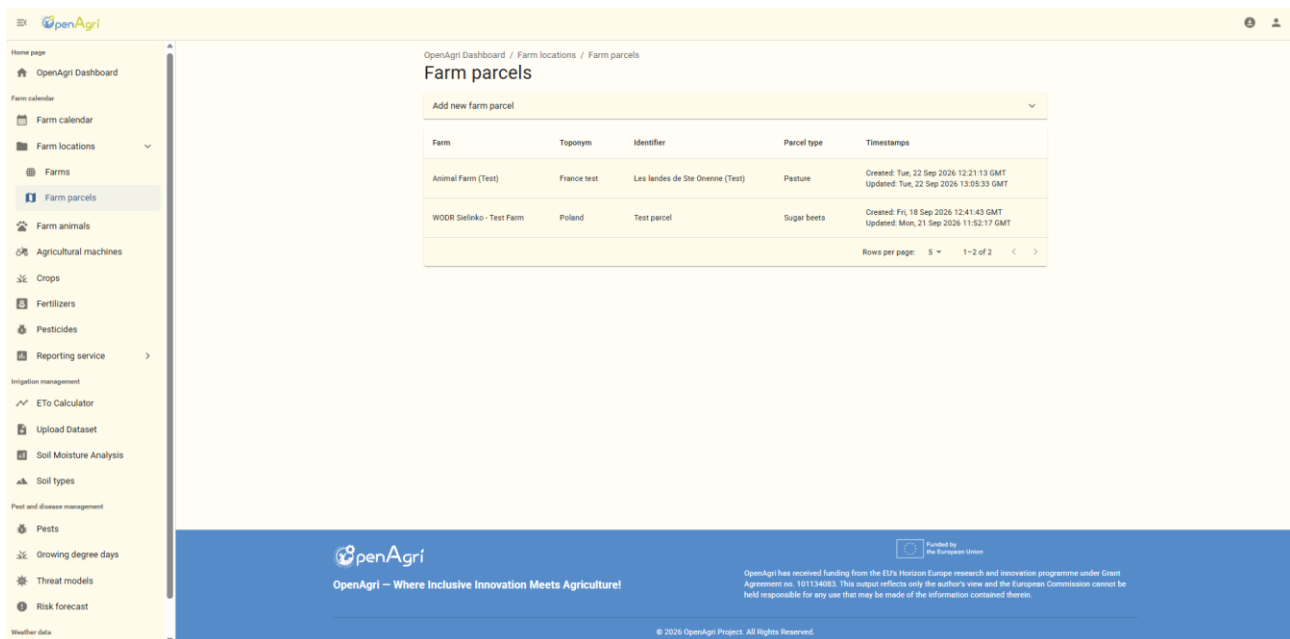


Figure 6. Farm parcels page

2.6.1. Purpose

Manages parcel-level information used by farm records, calculations, and reports.

2.6.2. Functional description

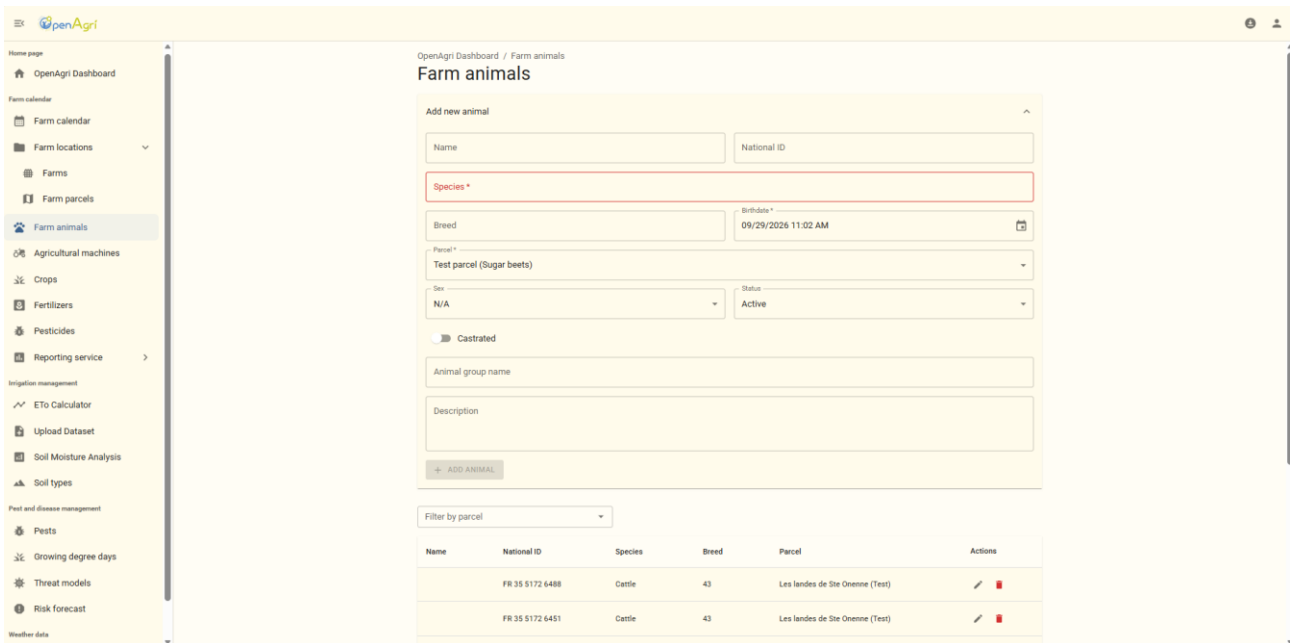
Farm parcels represent the spatial and agronomic units on which activities, crops, irrigation events, pest risks, and reports are recorded. The page displays parcel name, topology or location-related information, identifier, parcel crop, and timespan metadata.

Parcels are a key link between the farm record system and decision-support services. Selecting a parcel allows irrigation, ETo, soil moisture, pest, disease, and weather views to be interpreted for the correct field context. The service design supports semantic parcel representation and integration with other components, including geographical identifiers and OCSM-compatible data exchange.

2.6.3. User actions

- Review the list of parcels associated with a farm.
- Add new farm parcel records when new fields, plots, or areas are introduced.
- Use parcel records as filters for calendar, reporting, irrigation, weather, and pest/disease pages.

2.7. Farm Animals



Name	National ID	Species	Breed	Parcel	Actions
	FR 35 5172 6488	Cattle	43	Les landes de Ste Onenne (Test)	 
	FR 35 5172 6451	Cattle	43	Les landes de Ste Onenne (Test)	 

Figure 7. Farm animals page

2.7.1. Purpose

The Farm animals page allows users to register, review and manage livestock records associated with a selected farm parcel.

2.7.2. Functional description

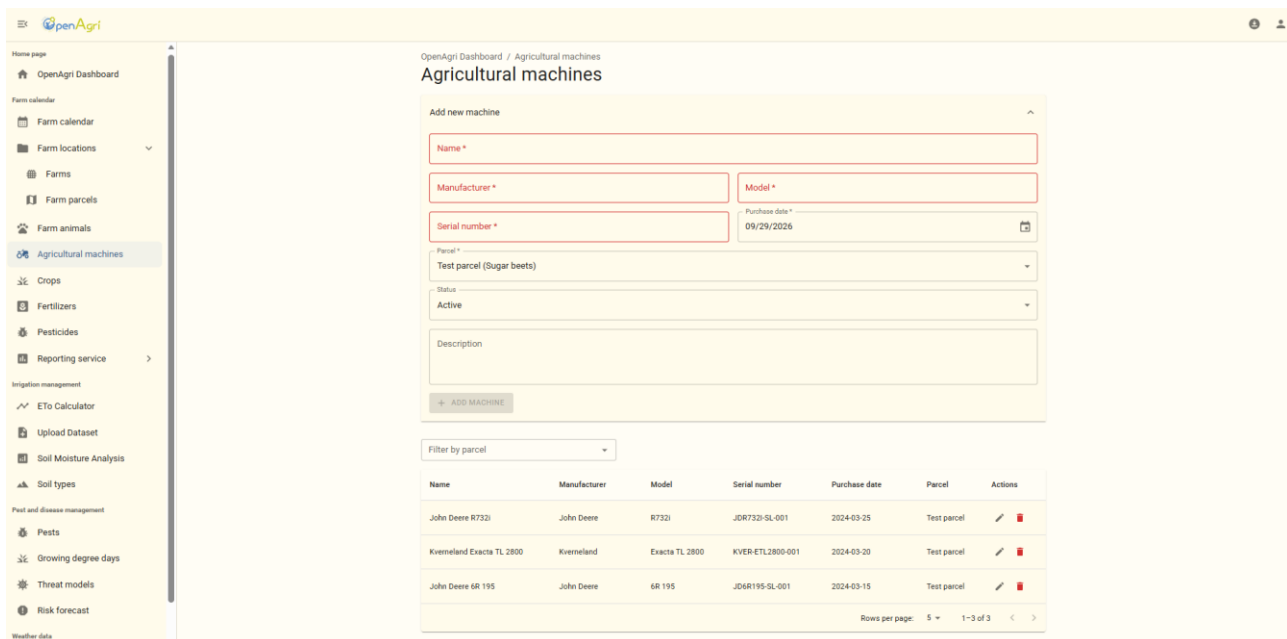
This page provides a structured form for adding a new animal to the OpenAgri platform. The form captures the main identification and classification data required for livestock management, including animal name, national ID, species, breed, birthdate, parcel, sex, status, castration status, animal group name and description.

Users can filter animals by parcel and review key information such as name, national ID, species, breed and assigned parcel.

2.7.3. User actions

- Add a new animal by completing the registration form and selecting the relevant parcel.
- Enter identification details such as name and national ID.
- Define biological and management attributes, including species, breed, birthdate, sex, status and castration status.
- Assign the animal to an animal group and add a free-text description when needed.
- Filter the animal list by parcel to focus on livestock connected to a specific farm location.
- Edit existing animal records using the edit action.
- Delete animal records using the delete action.

2.8. Agricultural machines



OpenAgri Dashboard / Agricultural machines

Agricultural machines

Add new machine

Name *

Manufacturer *

Model *

Serial number *

Purchase date * 09/29/2026

Parcel * Test parcel (Sugar beets)

Status Active

Description

+ ADD MACHINE

Filter by parcel

Name	Manufacturer	Model	Serial number	Purchase date	Parcel	Actions
John Deere R732i	John Deere	R732i	JD6732i-SL-001	2024-03-25	Test parcel	 
Kverneland Exakta TL 2800	Kverneland	Exakta TL 2800	KVER-ETL2800-001	2024-03-20	Test parcel	 
John Deere 6R 195	John Deere	6R 195	JD6R195-SL-001	2024-03-15	Test parcel	 

Rows per page: 5 1-3 of 3

Figure 8. Agricultural machines page

2.8.1. Purpose

The Agricultural machines page allows users to register, review and manage farm machinery records associated with selected farm parcels.

2.8.2. Functional description

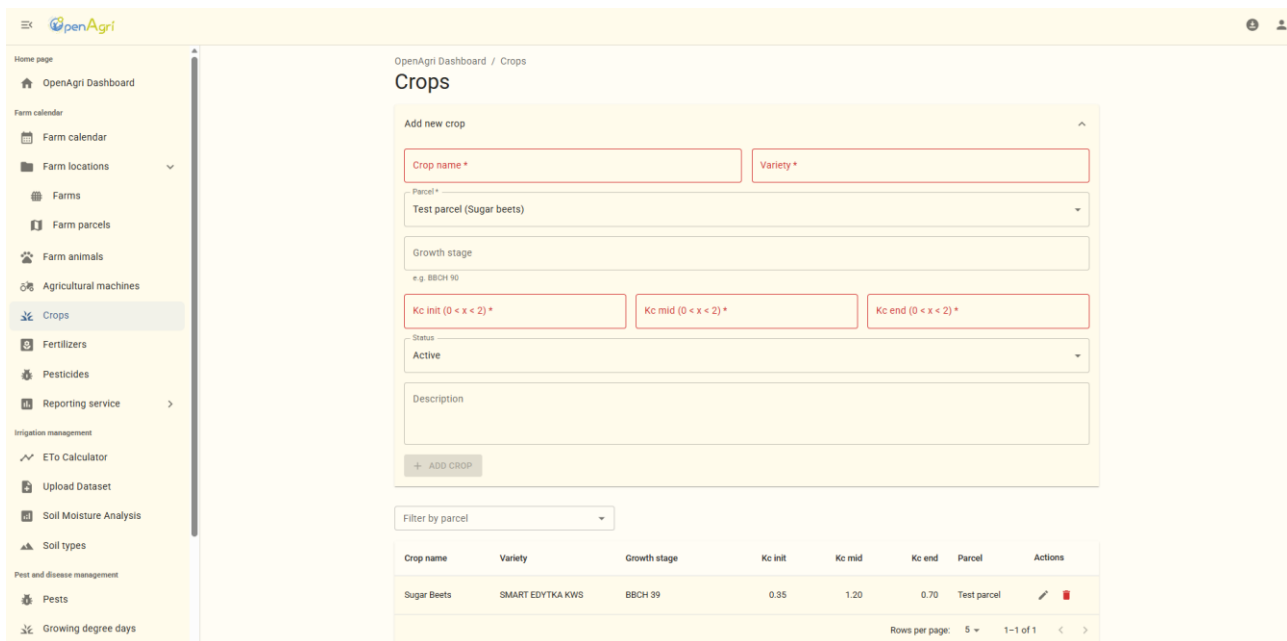
The form captures the main technical and administrative information required for machinery management, including machine name, manufacturer, model, serial number, purchase date, assigned parcel, operational status and description. Mandatory fields, such as name, manufacturer, model, serial number, purchase date and parcel, are visually highlighted to guide the user during data entry.

Users can filter machines by parcel and review key information such as name, manufacturer, model, serial number, purchase date and assigned parcel. Each record includes an actions area for managing existing entries, while pagination controls support navigation through larger machinery inventories.

2.8.3. User actions

- Add a new agricultural machine by completing the registration form.
- Enter identification and technical details such as name, manufacturer, model and serial number.
- Select the purchase date using the date field.
- Assign the machine to the relevant farm parcel.
- Set the machine status, such as active or inactive.
- Add a free-text description with additional operational or maintenance information.
- Filter the machinery list by parcel to focus on equipment connected to a specific farm location.
- Manage existing machinery records using the available action controls.

2.9. Crops



OpenAgri Dashboard / Crops

Crops

Add new crop

Crop name * Variety *

Parcel *

Growth stage

Kc init (0 < x < 2) * Kc mid (0 < x < 2) * Kc end (0 < x < 2) *

Status

Description

+ ADD CROP

Filter by parcel

Crop name	Variety	Growth stage	Kc init	Kc mid	Kc end	Parcel	Actions
Sugar Beets	SMART EDYTKA KWS	BBCH 99	0.35	1.20	0.70	Test parcel	 

Rows per page: 5 1-1 of 1

Figure 9. Crop types form

2.9.1. Purpose

The Crops page allows users to register and manage crop records within the Farm Calendar, linking crop information to farm parcels and supporting both field activity tracking and evapotranspiration calculations.

2.9.2. Functional description

This page provides a structured crop catalogue as part of the Farm Calendar. Users can register crops by defining the crop name, variety, related parcel, growth stage, status and description. The crop record is linked to a specific farm parcel, allowing the platform to connect crop information with calendar activities such as irrigation, fertilization, pesticide treatments and field observations.

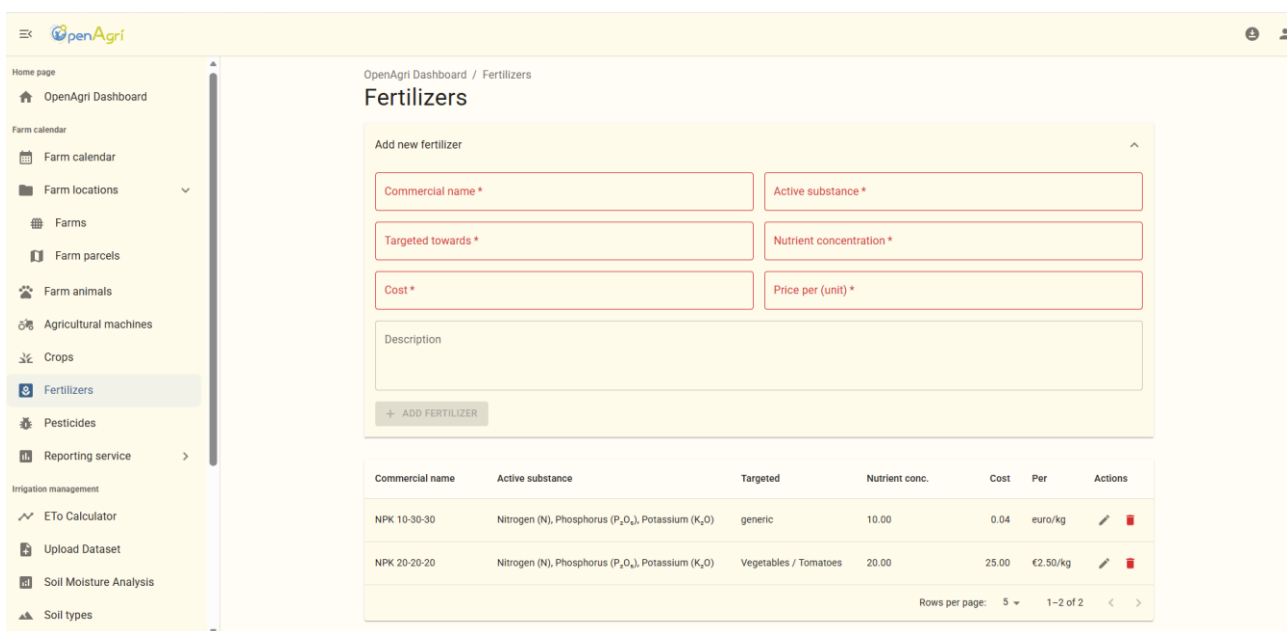
The page also stores crop coefficient values for the initial, mid-season and end growth stages. These Kc values are used by the evapotranspiration calculation service to adapt reference evapotranspiration values to the selected crop and growth stage. In this way, the Crops page supports both farm-record management in the Farm Calendar and more precise irrigation decision support through crop-specific evapotranspiration calculations.

The crop coefficient K_c is a crop-specific multiplier that adjusts baseline reference evapotranspiration (ET_0 - the water loss from a standardized well-watered grass surface) to determine the actual crop evapotranspiration ($ET_c = ET_0 \times K_c$). Because water consumption varies dramatically as a plant develops, having three distinct growth stages (Kc init for early growth/germination with low water demand, Kc mid for full canopy peak consumption, and Kc end for late-stage maturity/harvesting) allows the system to accurately scale daily weather data based on the crop's precise lifecycle phase, delivering accurate irrigation and crop water requirement estimations.

Crop Name	Kc init (Initial)	Crop Name	Kc init (Initial)
Maize (Grain Corn)	0.30	1.15	0.50
Wheat (Spring)	0.30	1.15	0.25
Potato	0.50	1.15	0.75
Tomato	0.60	1.15	0.80
Sugar Beet	0.35	1.20	0.70
Soybean	0.40	1.15	0.50
Cotton	0.35	1.15	0.50
Rice (Paddy)	01.05	1.20	0.75
Grapes (Table/Wine)	0.30	0.85	0.45
Alfalfa (Hay)	0.40	1.20	1.15

Above table showing frequently used crops along with their standard single crop coefficient (Kc) values according to the widely referenced FAO Irrigation and Drainage Paper No. 56.

2.10. Fertilizers



OpenAgri Dashboard / Fertilizers

Fertilizers

Add new fertilizer

Commercial name * Active substance *

Targeted towards * Nutrient concentration *

Cost * Price per (unit) *

Description

+ ADD FERTILIZER

Commercial name	Active substance	Targeted	Nutrient conc.	Cost	Per	Actions
NPK 10-30-30	Nitrogen (N), Phosphorus (P ₂ O ₅), Potassium (K ₂ O)	generic	10.00	0.04	euro/kg	 
NPK 20-20-20	Nitrogen (N), Phosphorus (P ₂ O ₅), Potassium (K ₂ O)	Vegetables / Tomatoes	20.00	25.00	€2.50/kg	 

Rows per page: 5 1-2 of 2

Figure 10. Fertilizers page

2.10.1 Purpose

The Fertilizers page allows users to register, review and manage fertilizer products used in farm operations.

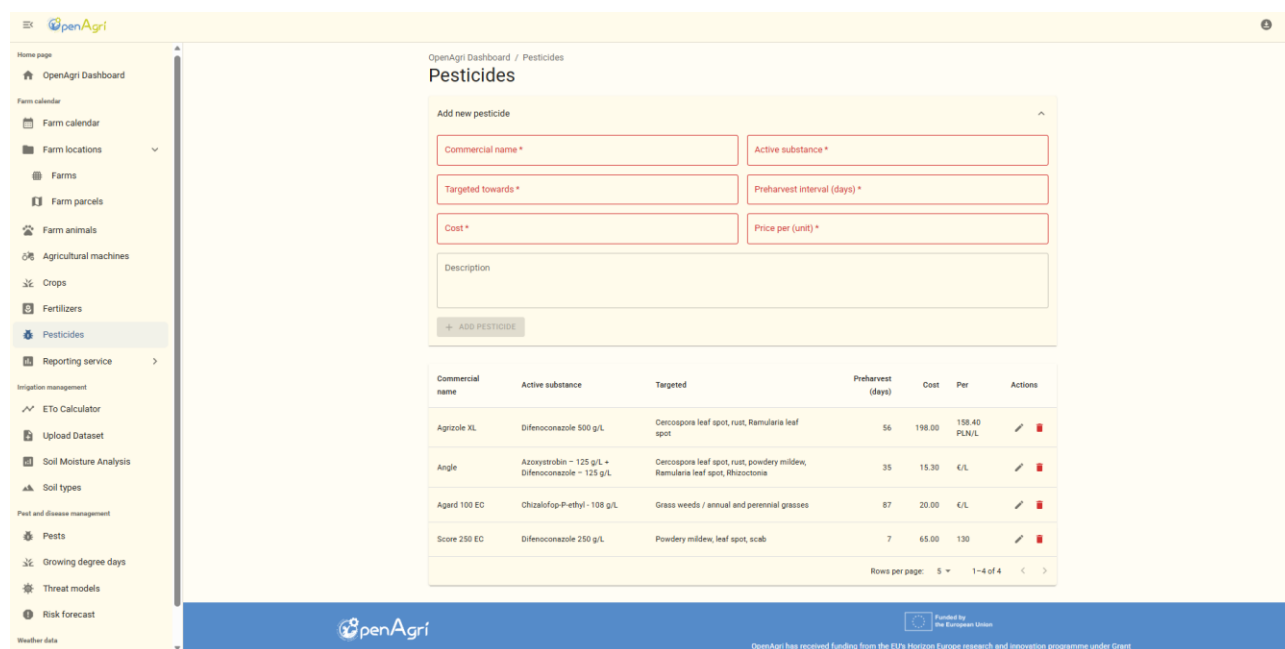
2.10.2 Functional description

This page provides a structured form for adding a new fertilizer to the OpenAgri platform. The form captures the key product information required for fertilizer management, including commercial name, active substance, targeted crop or use case, nutrient concentration, cost, price per unit and description. Mandatory

fields are visually highlighted to guide the user and ensure that the fertilizer record contains the minimum information needed for later use in reporting and field-operation records.

Below the registration form, the page displays the existing fertilizer records in a table. Users can review the commercial name, active substance, target use, nutrient concentration, cost and unit price. Each listed fertilizer includes action buttons for editing or deleting the record, allowing users to keep the fertilizer catalogue accurate and up to date.

2.11. Pesticides



Commercial name	Active substance	Targeted	Preharvest (days)	Cost	Per	Actions
Agrazole XL	Difenoconazole 500 g/L	Cercospora leaf spot, rust, Ramularia leaf spot	56	198.00	198.40 PLN/L	 
Angle	Azoxystrobin - 125 g/L + Difenoconazole - 125 g/L	Cercospora leaf spot, rust, powdery mildew, Ramularia leaf spot, Rhizoctonia	35	15.30	€/L	 
Agard 100 EC	Chlazifop-P-ethyl - 108 g/L	Grass weeds / annual and perennial grasses	87	20.00	€/L	 
Score 250 EC	Difenoconazole 250 g/L	Powdery mildew, leaf spot, scab	7	65.00	130	 

Rows per page: 5 1-4 of 4

Figure 11. Pesticides page

2.11.1 Purpose

The Pesticides page allows users to register, review and manage pesticide products that can later be used in Farm Calendar pesticide treatment activities.

2.11.2 Functional description

This page provides a structured catalogue for pesticide products used on farm parcels. Users can add a new pesticide by entering its commercial name, active substance, target pest or disease, preharvest interval, cost, price per unit and an optional description. Mandatory fields are visually highlighted to ensure that each pesticide record contains the key information required for safe and traceable crop-protection documentation.

In the Farm Calendar context, the pesticide catalogue supports the registration of pesticide treatment operations. When a user records a crop-protection activity, the pesticide product can be selected from this predefined list, helping to standardise activity records and improve the quality of reporting. The table below the form displays existing pesticide records with key product and regulatory information, including the preharvest interval in days, which is important for planning treatments and documenting compliance before harvest.

3. Reporting Service

3.1. Compost operations

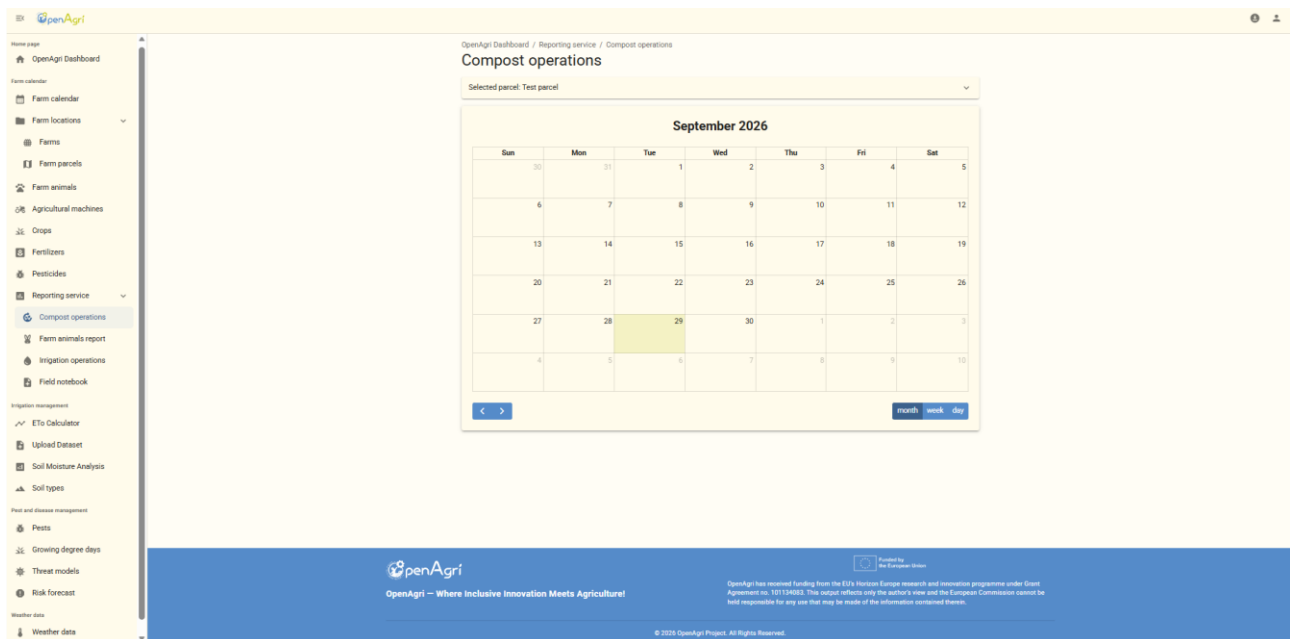


Figure 12. Compost operations page

3.1.1. Purpose

Supports the review and reporting of compost-related farm operations.

3.1.2. Functional description

The Compost operations page presents compost activities in a calendar-style view for the selected parcel. It is designed for workflows where compost production, raw material additions, and related operations must be recorded and reported.

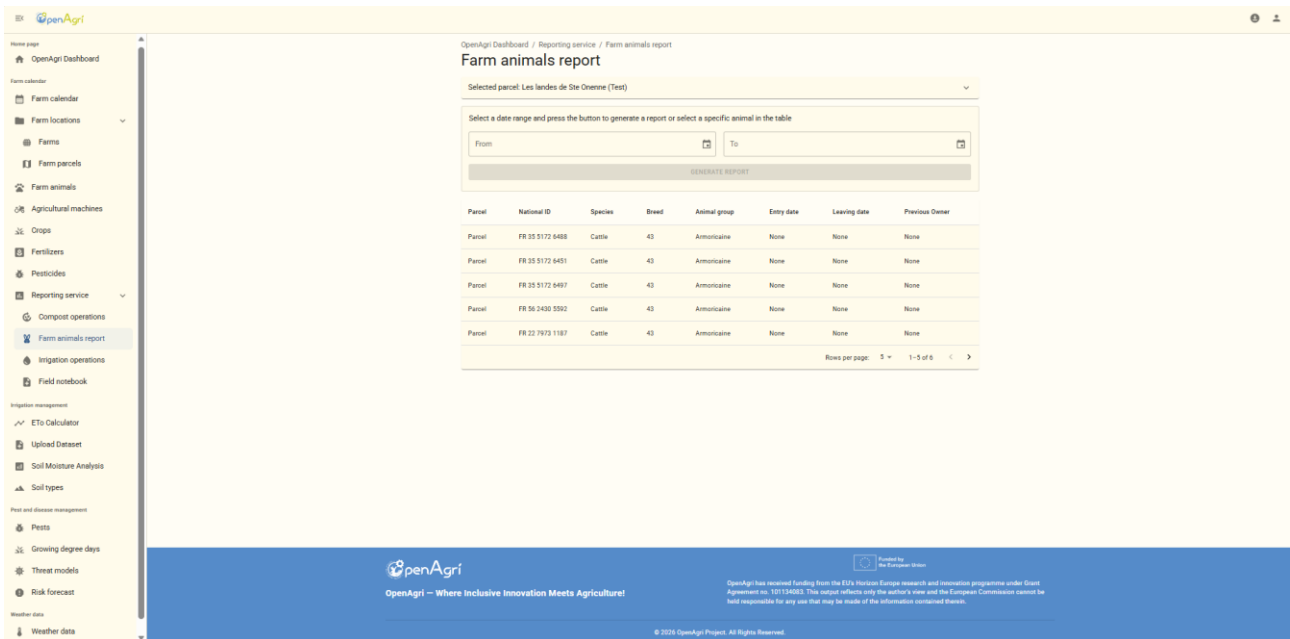
Within the OpenAgri service suite, compost operations are linked to the Farm Calendar and Reporting Service. They support the generation of domain-specific reports and enable the same data to be reused in audit, advisory, or certification processes.

The functionality is particularly relevant to OpenAgri use cases dealing with organic compost production and the documentation of applied farming practices.

3.1.3. User actions

- Select the parcel whose compost operations should be reviewed.
- Use the calendar to inspect operations by date.
- Use the underlying activity records as input for compost production reporting.

3.2. Farm animals report



OpenAgri Dashboard / Reporting service / Farm animals report

Farm animals report

Selected parcel: Les landes de Ste Onenne (Test)

Select a date range and press the button to generate a report or select a specific animal in the table

From To

[GENERATE REPORT](#)

Parcel	National ID	Species	Breed	Animal group	Entry date	Leaving date	Previous Owner
Parcel	FR 35 5172 9488	Cattle	43	Amurcaine	None	None	None
Parcel	FR 35 5172 9451	Cattle	43	Amurcaine	None	None	None
Parcel	FR 35 5172 9497	Cattle	43	Amurcaine	None	None	None
Parcel	FR 56 2430 5592	Cattle	43	Amurcaine	None	None	None
Parcel	FR 22 7873 1187	Cattle	43	Amurcaine	None	None	None

Rows per page: 5 1-5 of 5

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Figure 13. Farm animals report..

3.2.1. Purpose

Provides reporting functionality for livestock-related farm records.

3.2.2. Functional description

The Farm animals page allows the user to generate a livestock report for a selected date range or by selecting a specific animal from the table. The table supports structured animal information such as parcel, national ID, species, breed, animal group, key date, location, and producer.

The Reporting Service is designed to help farmers address reporting obligations by extracting domain-specific reports from already-recorded data. Livestock reports are one of the supported report types in the OpenAgri software suite.

This view contributes to the 'Collect Data Once, Use Many Times' principle: animal records created and maintained in the platform can be reused for internal management, advisory work, or formal reports.

3.2.3. User actions

- Select a parcel and reporting period.
- Generate a report for all animals in the selected period or select one animal from the table.
- Review livestock attributes before exporting or using the report.

3.3. Irrigation operations

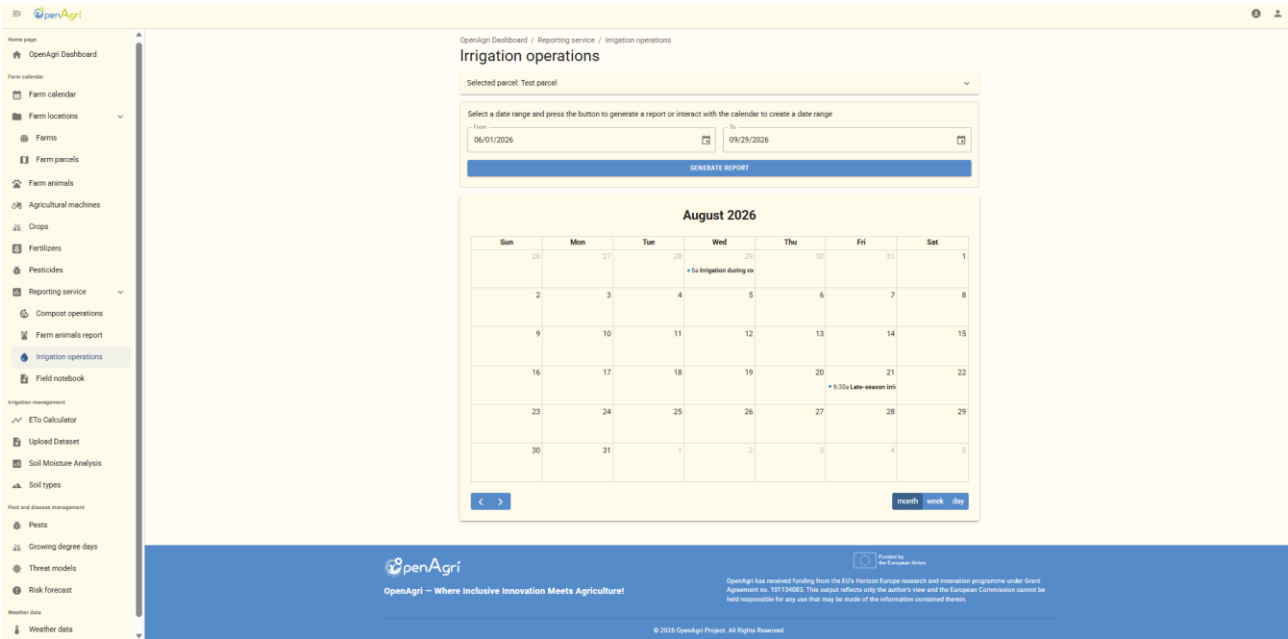


Figure 14. Irrigation operations page



Figure 15. Irrigation operations report

3.3.1. Purpose

Generates and reviews reports for irrigation operations performed on a parcel.

3.3.2. Functional description

The Irrigation operations page combines a date range selector, parcel selector, report generation button, and calendar view. It helps users review where and when irrigation activities were recorded.

Irrigation operation reports can be used for farm management, advisory analysis, compliance documentation, and comparison with soil moisture or weather data. They complement the Irrigation Management service, which analyses the efficiency and effects of irrigation.

The page is part of the Reporting Service, which automatically extracts relevant reports from operational records and structured agricultural datasets.

3.3.3. User actions

- Choose the parcel and the required reporting period.
- Generate the irrigation report.
- Inspect irrigation activities in the calendar and use them for decision-making or documentation.

3.4. Field notebook

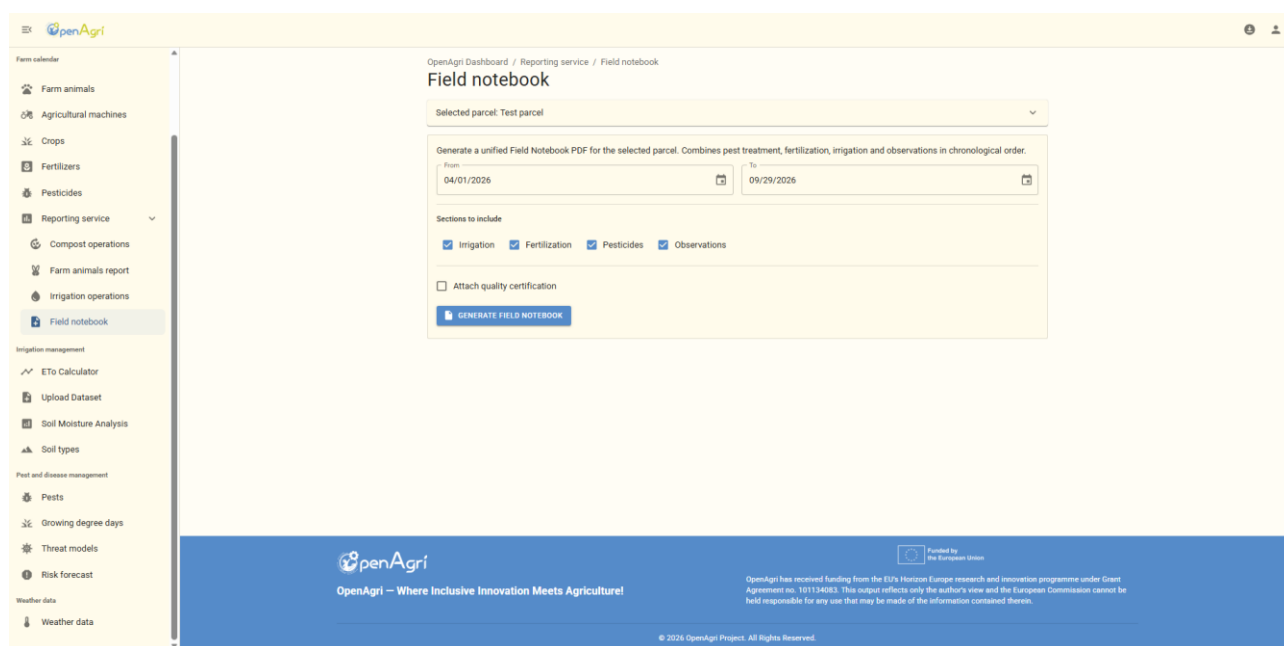


Figure 16. Field notebook page

3.4.1. Purpose

The Field notebook page allows users to generate a unified Field Notebook PDF for the selected parcel and reporting period.

3.4.2. Functional description

This functionality generates a Field Notebook report aligned with European farm diary and field-record requirements. The report combines farm and parcel information, crop-related data and recorded agronomic

activities such as pest treatments, fertilization and irrigation, together with observations, in chronological order.

Users can select the parcel, define the reporting period, choose which sections to include, and optionally attach quality certification information. The generated PDF provides a structured and traceable overview of field operations, supporting reporting, advisory work, certification processes and farm-management documentation.



Field Notebook

Generated: 29/09/2026 | Reporting period: 01/05/2026 – 29/09/2026

1. Farm & Parcel Information

Parcel Location: Country: Polska | City: | Postcode: 64-330

Parcel Identifier: Test parcel

Farm Location: Name: WODR Sielinko - Test Farm | Municipality: Opalenica

Administrator: OpenAgri Project

Contact Person: OpenAgri Project

Farm vat: TEMP-9370C0

Farm Description: Sugar beets (3ha) test fields with irrigation



Crops

Name	Description	Status	Growth Stage	Species	Variety
------	-------------	--------	--------------	---------	---------

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Name	Description	Status	Growth Stage	Species	Variety
Sugar Beets	Sugar beet crop grown under the CONVISIO® SMART production system. The crop is monitored throughout the growing season for irrigation requirements, crop development, pests and diseases.	1	BBCH 39	Sugar Beets	SMART EDYTKA KWS

Figure 17. Field notebook

4. Irrigation Management

4.1. Eto Calculator

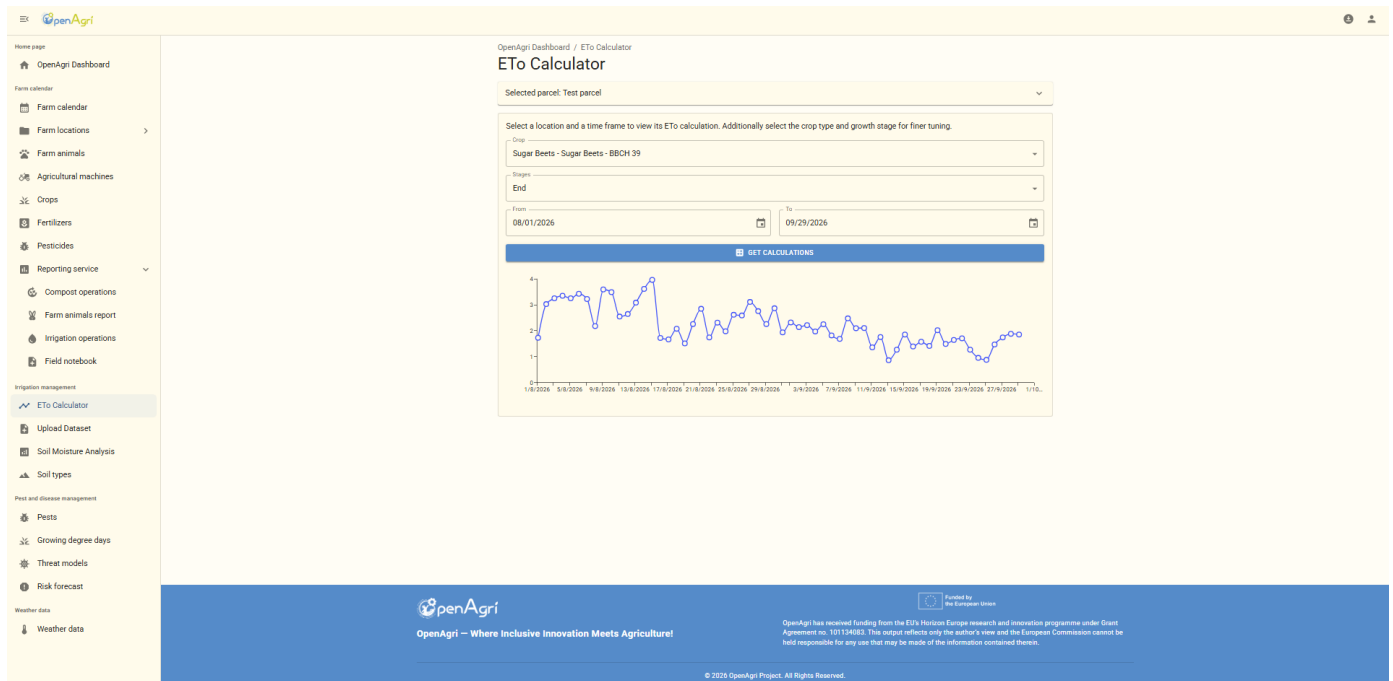


Figure 18. ETo Calculator

4.1.1. Purpose

Calculates reference evapotranspiration (ET_o) to support irrigation planning.

4.1.2. Functional description

The ETo Calculator is part of the OpenAgri Irrigation Management service. It estimates reference evapotranspiration using meteorological data and the FAO Penman-Monteith method. ET_o indicates how much water is transferred from the soil and vegetation into the atmosphere.

The page allows the user to select a parcel, location, crop, crop stage, and period, then run the calculation. The resulting chart displays ET_o values over time and can be used as an input to irrigation decisions.

The calculation can reuse parcel information from the Farm Calendar and weather parameters from the Weather Data Service.

After the location is created or selected, the user chooses a start date and end date. The service then retrieves ET_o calculations for that location and returns a list of daily results.

Each returned record contains:

- Date – the day for which the calculation applies.
- Value – the daily ET_o value in millimetres per day.

The platform can display these values as a time-series chart. The horizontal axis represents dates, while the vertical axis represents daily ET_o. Individual points show daily values and the connecting line helps reveal trends in evaporative demand across the selected period.

4.1.3. How to Interpret the Chart

A rising ETo trend usually indicates that weather conditions are becoming more demanding for crops. Warm, dry, sunny or windy periods commonly lead to higher ETo values. A falling trend can indicate cooler, wetter or less demanding conditions.

The ETo graph should be interpreted together with other information available in the OpenAgri platform, including:

- soil moisture measurements;
- rainfall records;
- crop growth stage;
- irrigation history;
- local agronomic recommendations.

ETo alone is not an irrigation command. It represents a reference climatic indicator. The actual irrigation requirement depends on the crop, soil type, rooting depth, irrigation efficiency, rainfall and the current amount of water stored in the soil.

4.1.4. Operational Use

Farmers and advisers can use the ETo page to identify periods when irrigation demand is increasing, compare different weeks or months, and support irrigation scheduling decisions. For example, consistently high ETo values combined with declining soil-moisture readings can indicate that a parcel should be reviewed for a possible irrigation event.

By combining geographic parcel information, date-based calculations and visual trend analysis, the ETo page transforms weather-derived water-demand data into an accessible decision-support tool.

4.1.5. User actions

- Select the parcel and location.
- Choose crop and crop stage filters.
- Set the start and end date and run ETo calculation.
- Review the chart to understand water demand trends.

4.2. Upload Dataset

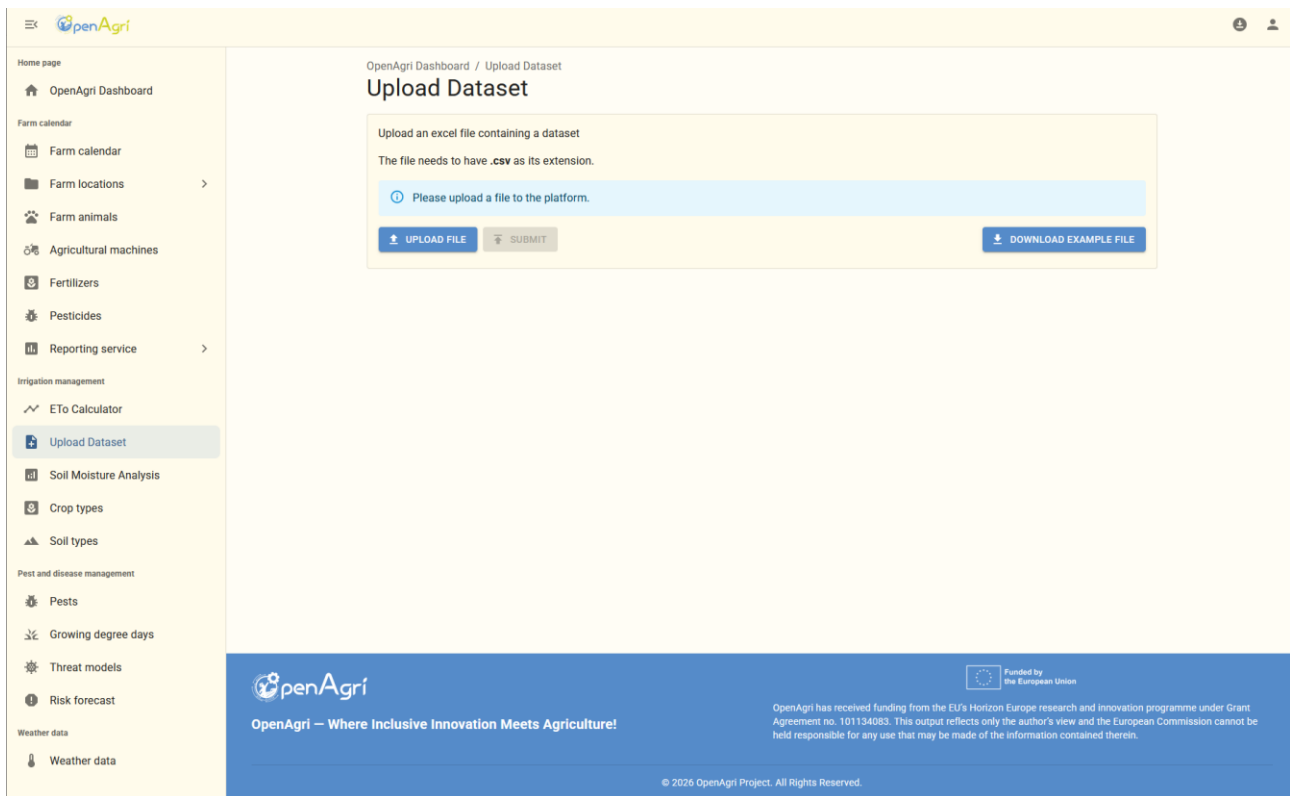


Figure 19. Upload Dataset

4.2.1. Purpose

Uploads soil moisture datasets for analysis by the Irrigation Management service.

4.2.2. Functional description

The Upload Dataset page is used to import CSV datasets containing soil moisture measurements and related weather information. The OpenAgri irrigation service supports datasets with timestamps, soil moisture values at multiple depths, rain, temperature, and humidity.

The service can process datasets even when some depth columns are missing, and it can use depth-specific weights to adapt the analysis to the sensor setup. This allows the same analytics engine to be reused across different field installations.

After upload, the dataset becomes available for soil moisture analysis and irrigation efficiency assessment.

4.2.3. User actions

- Select a local CSV file.
- Upload the file to the platform.
- Download a sample CSV format when needed.
- Continue to Soil Moisture Analysis after the upload is complete.

4.3. Soil Moisture Analysis

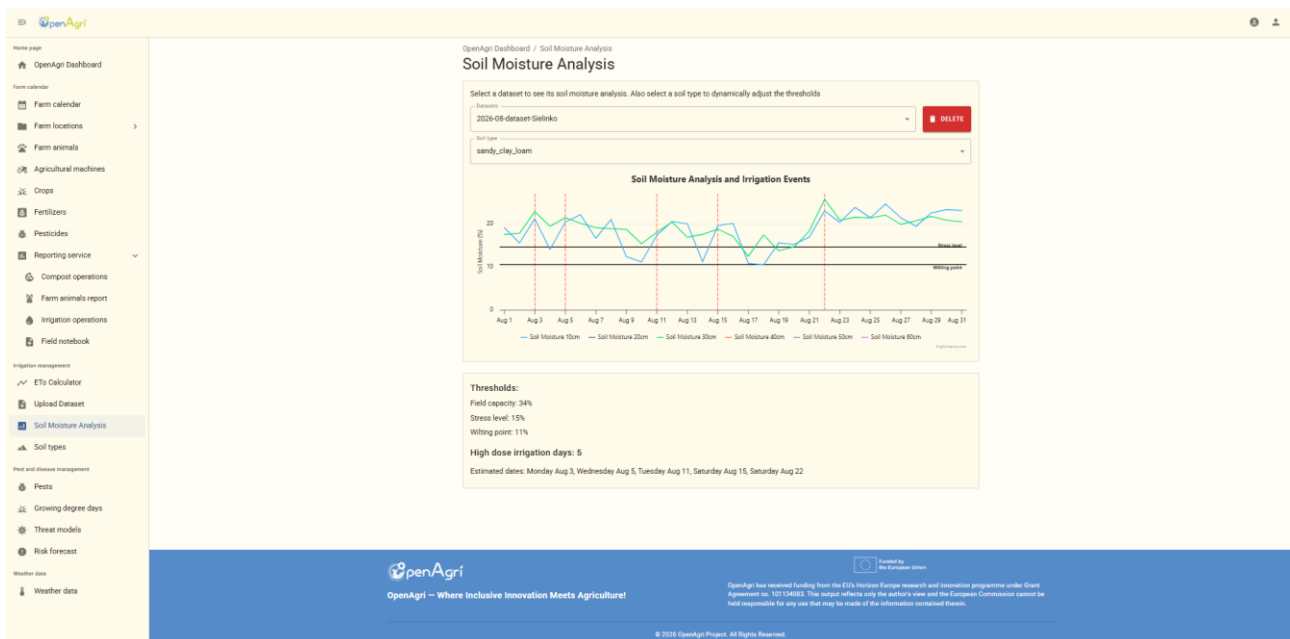


Figure 20. Soil Moisture Analysis

4.3.1. Purpose

Analyses soil moisture time-series data and identifies irrigation-related events and thresholds.

4.3.2. Functional description

The Soil Moisture Analysis page visualises soil moisture data together with irrigation and precipitation events. The service analyses time-series measurements to detect irrigation events, precipitation events, high-dose irrigation events, field capacity, stress level, saturation days, and stress days.

The chart helps users interpret sensor readings that would otherwise require expert knowledge. Vertical markers indicate irrigation or high-dose irrigation events, while summary indicators below the chart provide threshold and event counts.

The functionality supports irrigation planning by highlighting excessive irrigation, possible water stress, and soil saturation patterns for the selected dataset and soil type.

The input dataset may include soil-moisture readings from several sensor depths, for example 10 cm, 20 cm, 30 cm, 40 cm, 50 cm and 60 cm. The service normalises recognised column names and prepares the sensor data for analysis.

Rainfall, temperature and humidity are required because they provide important environmental context. Timestamps should use a consistent ISO-like format so that sensor observations and detected events can be aligned correctly over time.

When a soil-depth measurement is missing, the current implementation may fill that depth with zero values. This supports technical completeness of the dataset but should be carefully interpreted: a flat zero line may indicate missing source data rather than genuinely dry soil.

4.3.3. Detected Events

After the dataset is uploaded, the service analyses the time series and returns detected irrigation-related indicators. Two important outputs are:

- High-dose irrigation events
- Saturation days

The platform can display soil moisture lines for each depth and mark high-dose irrigation events with red dashed vertical lines. Saturation days can be displayed using blue dashed vertical lines.

4.3.4. Reading High-Dose Irrigation Events

After a high-dose irrigation marker, moisture values in shallow sensors, such as 10 cm or 20 cm, are expected to increase quickly. Deeper sensors may respond later, depending on soil structure, irrigation volume and infiltration speed.

A delayed or weak response at deeper layers can suggest that water is not reaching the deeper root zone, that infiltration is slow, or that the irrigation dose was insufficient for deeper soil layers.

If an irrigation event is marked but no moisture increase is visible, the user should check sensor calibration, measurement timestamps, irrigation timing and the relationship between irrigation volume and sensor sampling frequency.

4.3.5. Reading Saturation Days

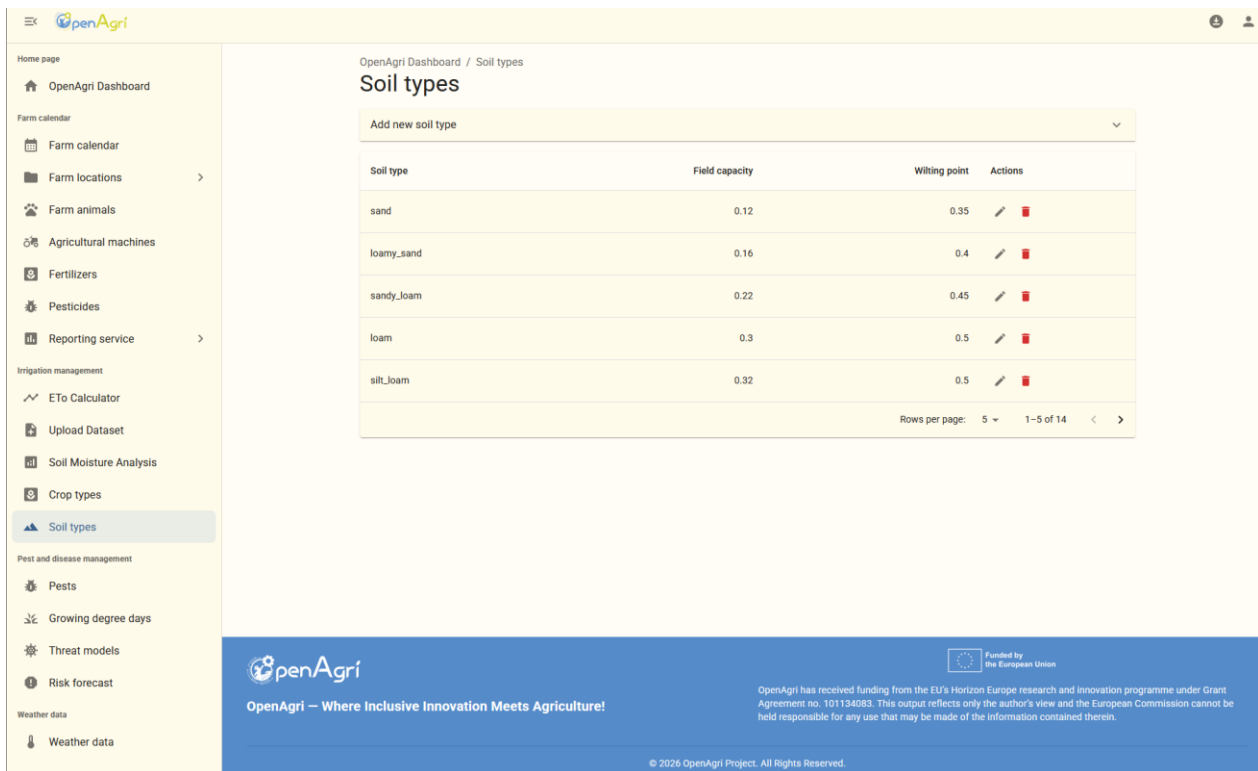
Repeated saturation markers may indicate excessive irrigation, poor drainage, compacted soil or a high water table. Saturation after an irrigation event is not always negative, but frequent or persistent saturation can increase the risk of nutrient leaching, root stress and unnecessary water use.

The Soil Moisture Analysis page should therefore be used as a practical tool for comparing irrigation actions with the actual behaviour of water in the soil profile, supporting more efficient and evidence-based irrigation decisions.

4.3.6. User actions

- Select a dataset uploaded to the platform.
- Choose soil type or relevant analysis settings.
- Review the chart and the calculated thresholds.
- Use high-dose irrigation days, field capacity, and stress indicators to adjust future irrigation plans.

4.4. Soil types



OpenAgri Dashboard / Soil types

Soil types

Add new soil type

Soil type	Field capacity	Wilting point	Actions
sand	0.12	0.35	 
loamy_sand	0.16	0.4	 
sandy_loam	0.22	0.45	 
loam	0.3	0.5	 
silt_loam	0.32	0.5	 

Rows per page: 5 1-5 of 14

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Figure 21. Soil Types form

4.4.1. Purpose

Users can manage baseline soil properties (field capacity and wilting point) that enable the system's soil moisture analysis engine to accurately compute water retention, establish plant stress thresholds, and evaluate irrigation efficiency.

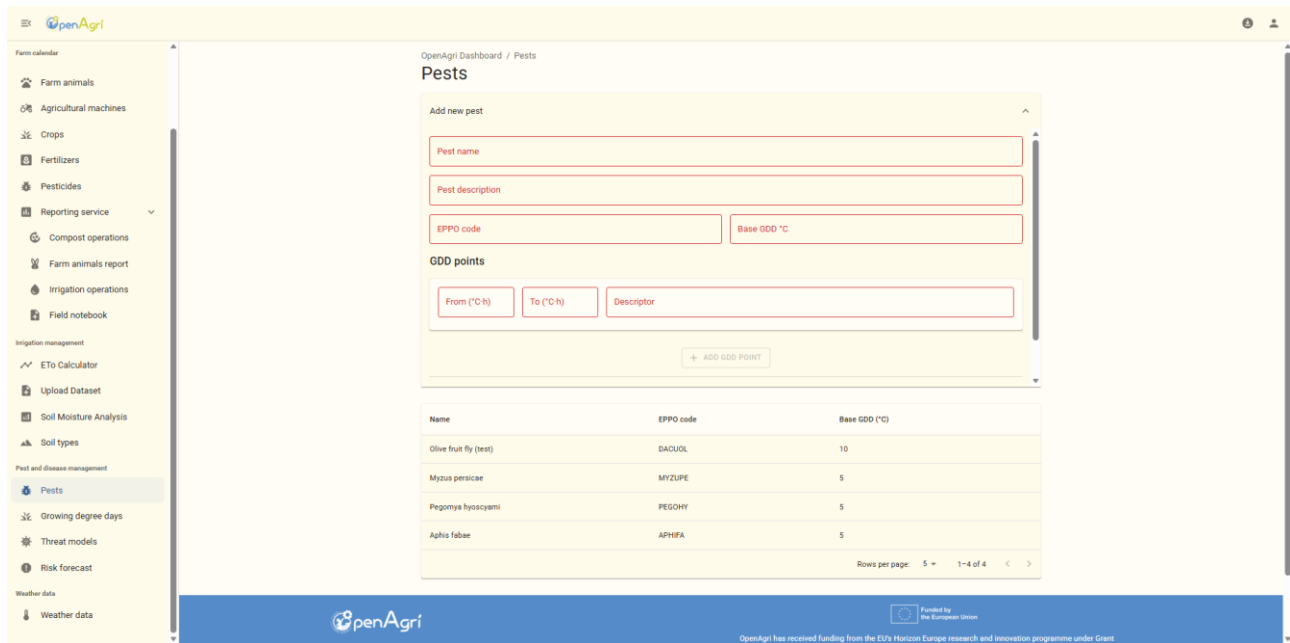
4.4.2. Functional description

"Soil types" page provides a form for adding new soil profile entries into the dashboard. Users enter a Soil type identifier alongside two crucial hydrological coefficients: Field capacity and Wilting point.

Field capacity and wilting point are essential soil-water parameters required by Soil Moisture Analysis engine to evaluate agricultural water availability and stress. Field Capacity represents the maximum volumetric water content the soil can retain against gravity after excess drainage, whereas Wilting Point represents the lower moisture threshold below which plants can no longer extract water, leading to crop stress. By combining these parameters with historical sensor telemetry and rain event data, the analysis engine automatically determines plant stress thresholds, quantifies available soil water, and assesses the overall efficiency of irrigation practices.

5. Pest and disease management

5.1. Pests



OpenAgri Dashboard / Pests

Pests

Add new pest

Pest name

Pest description

EPPO code

Base GDD °C

GDD points

From (°C·h) To (°C·h) Descriptor

+ ADD GDD POINT

Name	EPPO code	Base GDD (°C)
Olive fruit fly (test)	BACUOL	10
Myzus persicae	MYZUPE	5
Pegomya hyoscyami	PESOHY	5
Aphis fabae	APHIFA	5

Rows per page: 5 1-4 of 4

OpenAgri

OpenAgri has received funding from the EU's Horizon Europe research and innovation programme under Grant

Figure 22. Pests page

5.1.1. Purpose

Maintains pest and disease definitions and their environmental parameters.

5.1.2. Functional description

The Pests page is part of the Pest and Disease Management service. It supports the definition of pest/pathogen records and parameters that describe the environmental conditions under which infestation or disease pressure may occur.

The service is designed as a modular decision-support system that can combine weather conditions, crop type, growth stage, Growing Degree Days, and rule definitions to estimate pest and disease risk.

By managing pest definitions and thresholds in the platform, users can reuse the same knowledge in threat models and risk forecasts for selected parcels.

5.1.3. User actions

- Create or edit a pest/pathogen record.
- Define crop, humidity, temperature, rainfall, or other rule-related parameters.
- Review the table of existing pests and connect them to threat models.

5.2. Growing degree days

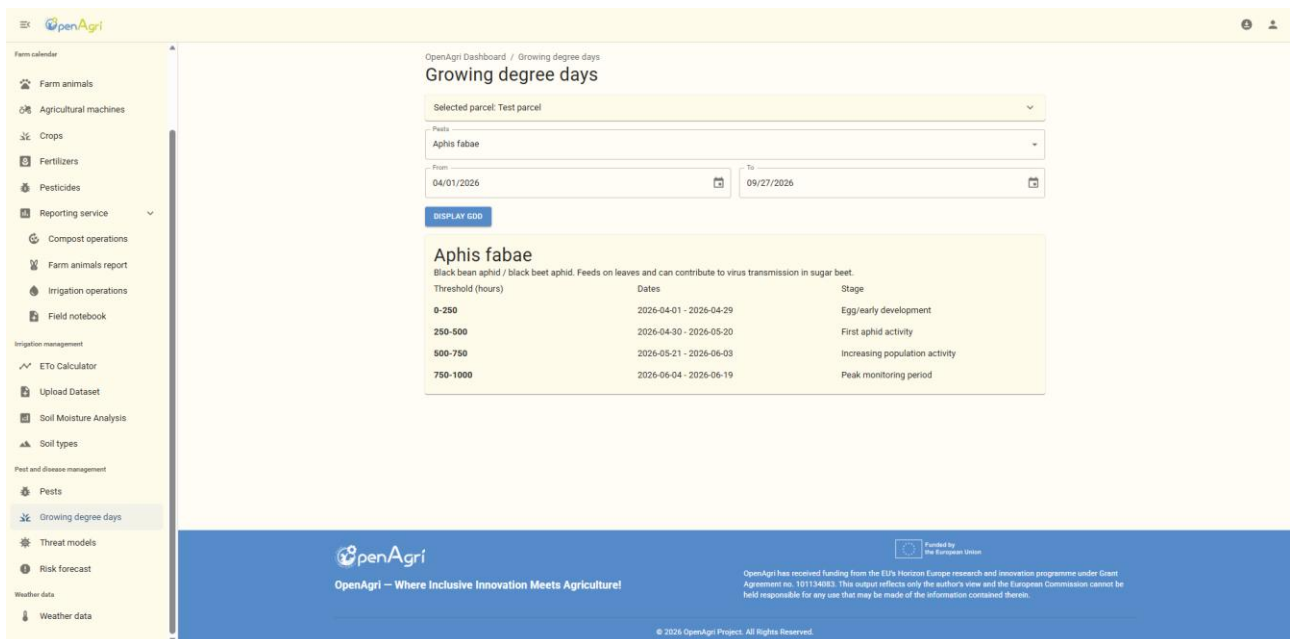


Figure 23. Growing Degree Days

5.2.1. Purpose

Calculates accumulated heat units to support crop and pest phenology assessment.

5.2.2. Functional description

Growing Degree Days (GDD) are used to estimate biological development based on temperature accumulation. In OpenAgri, GDD calculations support pest, disease, and crop growth decision-support workflows.

The page allows the user to select a parcel, model, and time period. The result section shows the selected model and accumulated GDD values, which can be used to interpret development stages or pest pressure.

For each day, the service retrieves minimum and maximum temperature data for the selected farm parcel. It then calculates daily heat accumulation using the following formula:

$$\text{GDD} = ((T_{\text{max}} + T_{\text{min}}) / 2) - \text{Base Temperature}$$

The base temperature is defined in the selected disease or pest model. It represents the lower temperature threshold below which meaningful development is not expected for the target organism.

When the calculated daily value is negative, the service sets it to zero. This prevents cold days from reducing the heat accumulation already achieved during earlier days.

The service calculates two values for each day:

- Daily GDD – heat accumulation for that specific day.
- Accumulated GDD – the running sum of daily GDD values from the beginning of the selected period.

5.2.3. Workflow in the Platform

The user first selects or defines a farm parcel and chooses a pest or disease model. Each model may include a name, EPPO code, base temperature, description and a list of GDD intervals associated with practical recommendations.

The user then selects a start date and end date. The platform retrieves the relevant weather data and produces a time series containing daily and accumulated GDD values. These results can be visualised as a graph, where the daily GDD line shows short-term temperature variation and the accumulated line shows seasonal development progress.

5.2.4. Decision Support Value

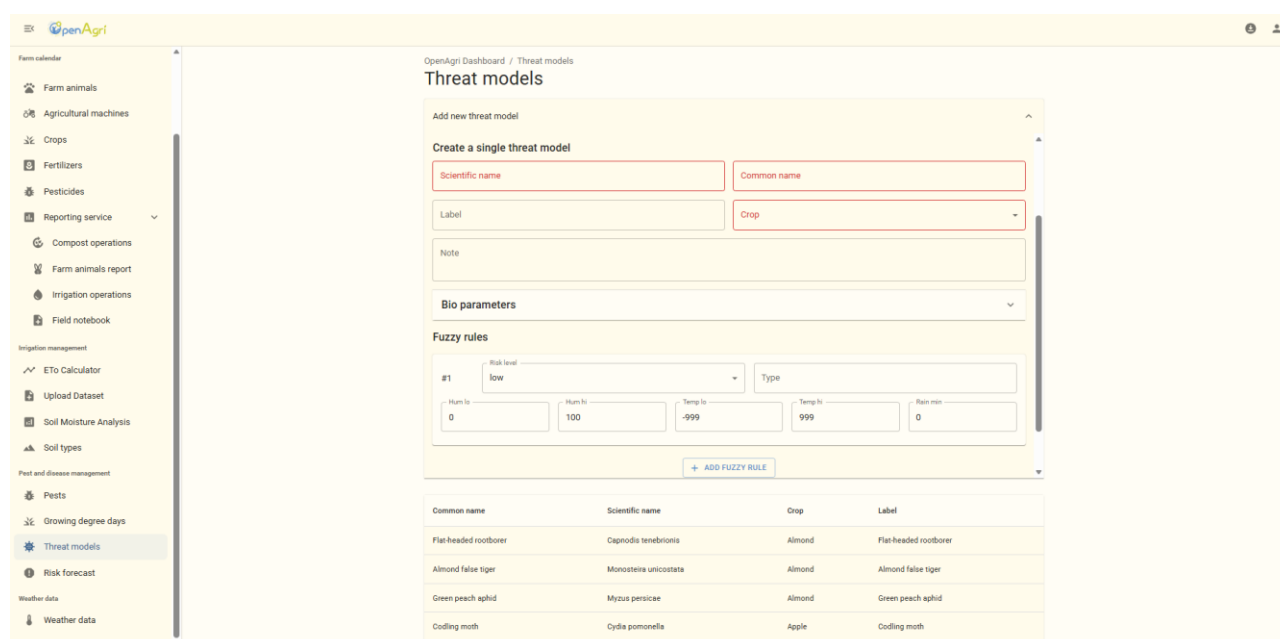
The accumulated GDD curve is compared with threshold ranges defined in the selected model. For example, a model can define periods such as:

- early development, when treatment is not required;
- a biologically optimal window for a biological control product;
- a window for conventional treatment;
- a later period in which the pest is no longer susceptible.

The purpose is not to prescribe treatment automatically. Instead, the GDD page provides a transparent temperature-based indicator that helps users schedule field inspections and identify the most suitable period for intervention.

5.2.5. Practical Interpretation

A slowly rising accumulated GDD curve indicates cool conditions and slower biological development. A steep increase indicates warmer conditions and faster development. By comparing the curve with model thresholds, the farmer can move from calendar-based planning to a more location-specific and data-driven approach.



Common name	Scientific name	Crop	Label
Flat-headed rootborer	Capnodis tenebrionis	Almond	Flat-headed rootborer
Almond false tiger	Monosteira uncostata	Almond	Almond false tiger
Green peach aphid	Myzus persicae	Almond	Green peach aphid
Codling moth	Cydia pomonella	Apple	Codling moth

Figure 24. Treat model definition.

5.2.6. Purpose

Creates reusable pest or disease threat models based on weather-related rules.

5.2.7. Functional description

The Threat models page supports the creation of a model that describes when a disease or pest risk should be considered relevant. The modal form includes basic model attributes, crop selection, rule parameters, and daily conditions.

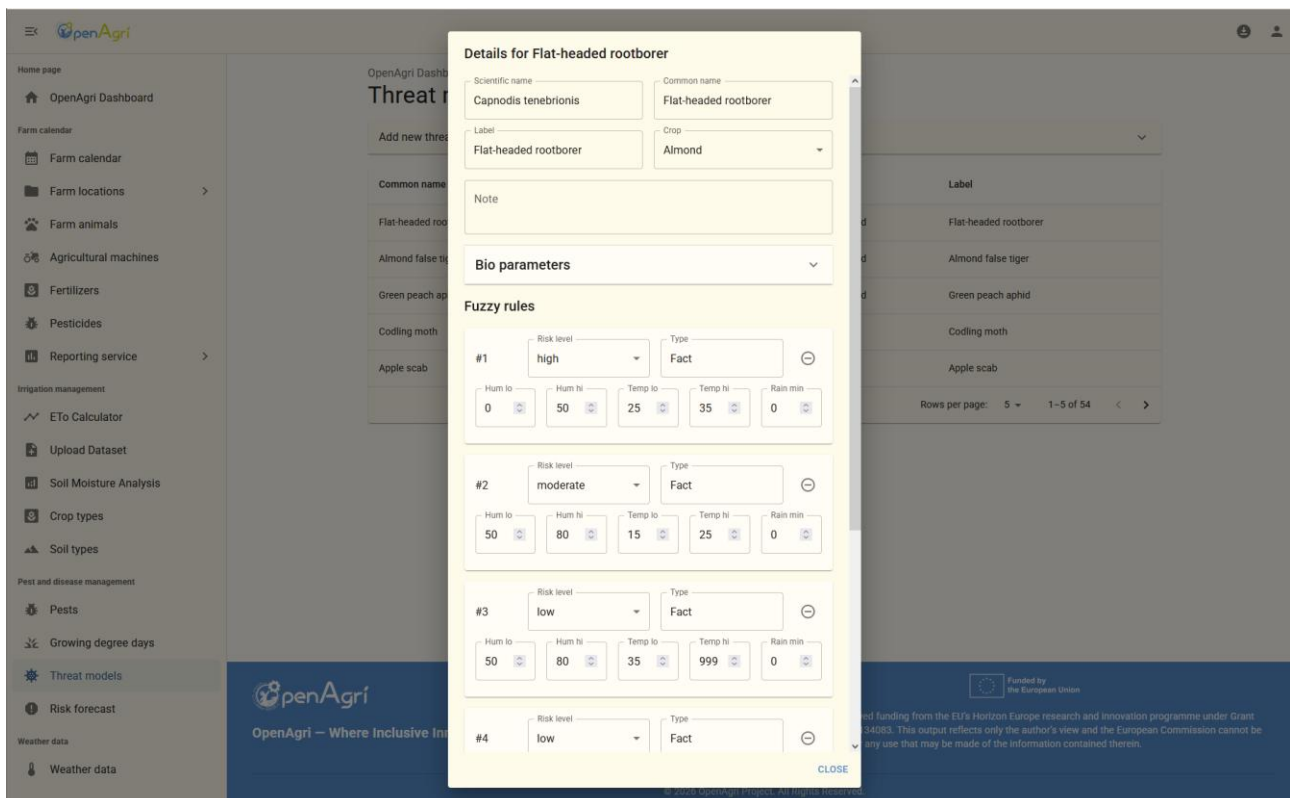
The Pest and Disease Management service allows users or other services to define rules, apply them to datasets, and retrieve whether the conditions were fulfilled over a period of time. Threat models formalise those rules so they can be reused in risk forecasts.

This approach allows the platform to start with models relevant to OpenAgri pilots and later extend to additional pests, diseases, crops, and local conditions.

5.2.8. User actions

- Create a new threat model or edit an existing one.
- Add model name, description, and crop.
- Define weather parameters and thresholds for the model.
- Save the model and use it on the Risk Forecast page.

5.3. Fuzzy Risk Engine

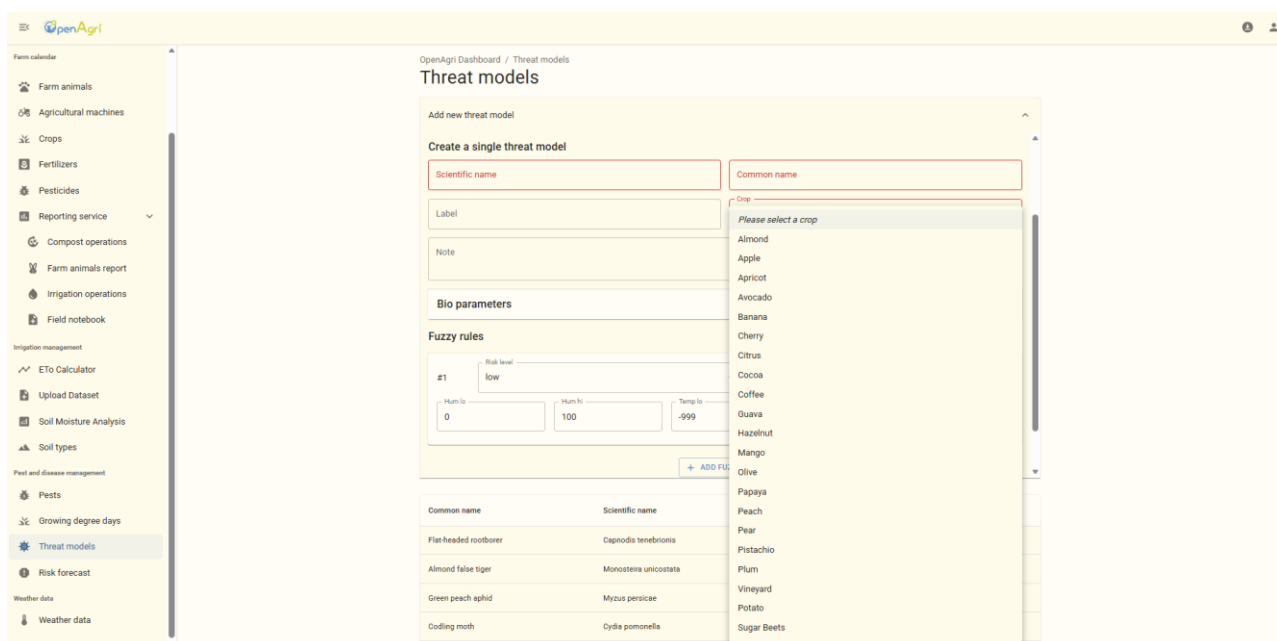


The screenshot shows the 'Details for Flat-headed rootborer' modal form. The form is divided into several sections:

- Scientific name:** Capnodis tenebrionis
- Common name:** Flat-headed rootborer
- Label:** Flat-headed rootborer
- Crop:** Almond
- Note:** (Empty text area)
- Bio parameters:** (Dropdown menu)
- Fuzzy rules:**
 - #1:** Risk level: high, Type: Fact. Parameters: Hum lo: 0, Hum hi: 50, Temp lo: 25, Temp hi: 35, Rain min: 0.
 - #2:** Risk level: moderate, Type: Fact. Parameters: Hum lo: 50, Hum hi: 80, Temp lo: 15, Temp hi: 25, Rain min: 0.
 - #3:** Risk level: low, Type: Fact. Parameters: Hum lo: 50, Hum hi: 80, Temp lo: 35, Temp hi: 999, Rain min: 0.
 - #4:** Risk level: low, Type: Fact. Parameters: (Not fully visible).

The background shows the OpenAgri Dashboard with various navigation options like Farm calendar, Farm locations, Farm animals, Agricultural machines, Fertilizers, Pesticides, Reporting service, Irrigation management, ETo Calculator, Upload Dataset, Soil Moisture Analysis, Crop types, Soil types, Pest and disease management, Pests, Growing degree days, Threat models, Risk forecast, and Weather data.

Figure 25. Fuzzy rules



OpenAgri Dashboard / Threat models

Threat models

Add new threat model

Create a single threat model

Scientific name

Common name

Label

Note

Bio parameters

Fuzzy rules

#1 Risk level: low

Hum to: 0 Hum to: 100 Temp to: -999

+ ADD RULE

Common name	Scientific name
Flat headed rootborer	Capnodis tenebrionis
Almond false tiger	Monosteira unicostata
Green peach aphid	Myzus persicae
Codling moth	Cydia pomonella

Crop: Please select a crop

- Almond
- Apple
- Apricot
- Avocado
- Banana
- Cherry
- Citrus
- Cocoa
- Coffee
- Guava
- Hazelnut
- Mango
- Olive
- Papaya
- Peach
- Pear
- Pistachio
- Plum
- Vineyard
- Potato
- Sugar Beets

Figure 26. Threat models

5.3.1. Purpose

The OpenAgri Fuzzy Risk Engine is an intelligent decision-support component within the Pest and Disease Management service. Its purpose is to convert weather observations, parcel context and crop-specific threat models into an understandable pest or disease risk assessment.

Unlike a strict threshold-based rule system, fuzzy logic is designed to work with gradual biological conditions. In real agricultural environments, there is rarely a sharp boundary between “safe” and “dangerous” weather. For example, temperature, humidity or rainfall conditions may be partly favourable for infection rather than simply true or false. The Fuzzy Risk Engine represents these gradual transitions using linguistic categories such as low, moderate, high or very high suitability.

5.3.2. How the Engine Works

The engine follows a Mamdani fuzzy inference approach. First, numerical environmental and contextual inputs are transformed into fuzzy membership values. A weather condition may therefore belong partly to more than one class. For example, a temperature can be considered both moderately favourable and highly favourable for a pathogen, with different degrees of confidence.

The engine then evaluates fuzzy rules defined for a selected crop-threat combination. These rules encode agronomic knowledge about the conditions that favour pest development, disease infection or epidemic spread. The relevant threat model is linked to a crop and can contain a scientific name, common name, label, explanatory note and a structured model definition.

After rule evaluation, the fuzzy outputs are combined and transformed into a final numerical risk score. The service returns a score in the range from 0 to 100 together with a qualitative risk classification:

- Low
- Moderate
- High
- Critical

5.3.3. User actions

- Select a parcel and GDD model.
- Set the date range for the calculation.
- Run the calculation and review accumulated values and model output.

5.4. Risk forecast

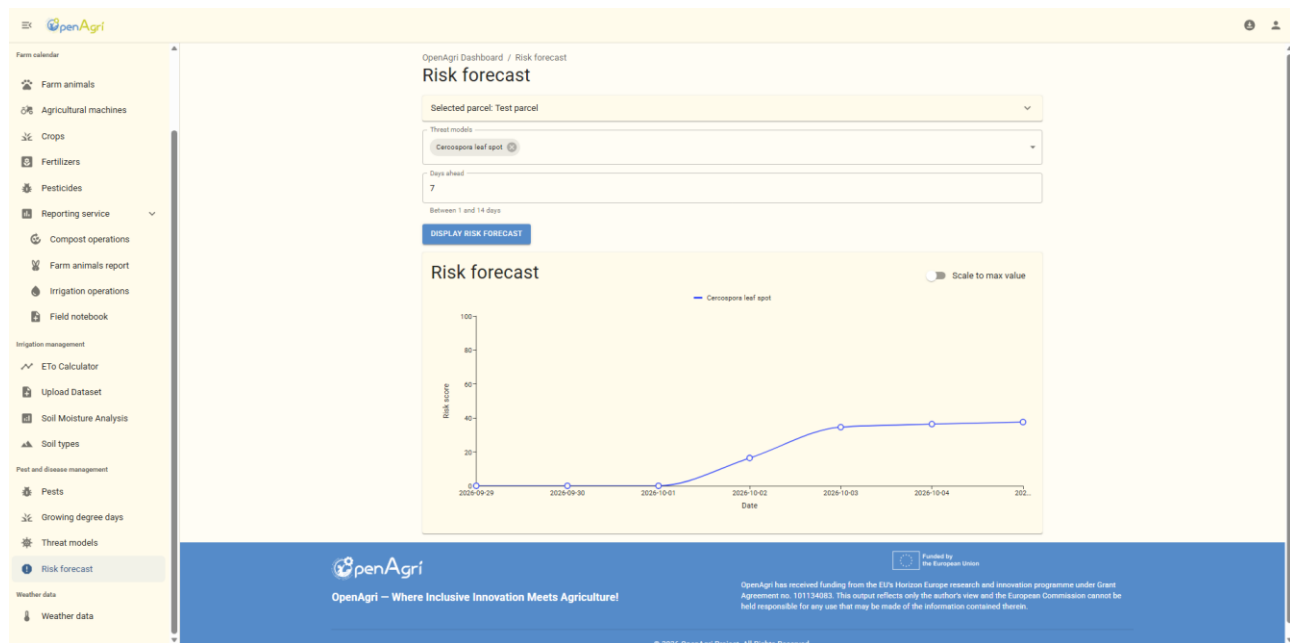


Figure 27. Risk forecast page

5.4.1. Purpose

Calculates and visualises the risk of pest or disease pressure for a selected parcel and model.

5.4.2. Functional description

The Risk forecast page combines parcel context, crop context, threat model selection, and weather-based calculations to estimate disease or pest risk. The output is displayed as a chart over time.

The underlying service can apply rule logic to selected datasets or weather data and return whether the rule conditions were met. This supports early warning and advisory decisions, for example when humidity, temperature, rainfall, or duration thresholds indicate elevated disease pressure.

The page is designed for practical decision support: it does not replace expert judgement, but it highlights periods that require attention, inspection, or preventive action.

5.4.3. User actions

- Select the parcel, threat model, and crop.
- Run risk forecast calculation.
- Inspect the risk curve and identify periods of higher probability.
- Use the result to plan scouting, advisory actions, or treatment decisions

6. Weather data

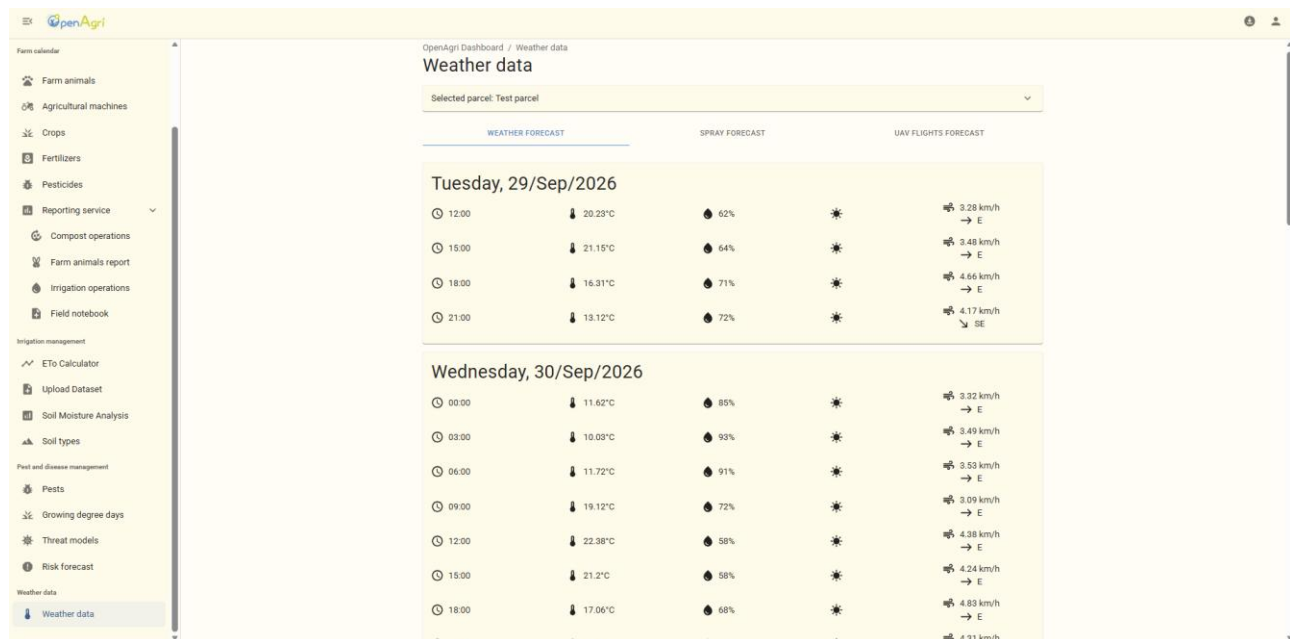


Figure 28. Weather data table

6.1.1. Purpose

Provides weather forecasts and agriculture-specific weather decision-support views.

6.1.2. Functional description

The Weather Data page presents forecast information for the selected parcel. The forecast view includes weather parameters such as time, temperature, humidity, precipitation or cloud indicators, and wind speed/direction.

The OpenAgri Weather Data Service integrates external weather providers and exposes forecast and historical data through a REST API and JSON-LD/OCSM-compatible outputs. It provides 5-day forecasts, up to one-month historical data, and indicators such as Temperature-Humidity Index (THI).

The page also includes tabs for spraying suitability and UAV flight suitability. These decision-support views evaluate whether weather conditions are appropriate for safe crop spraying or drone-based agricultural operations. Offline resilience is supported through local caching of recent historical data for configured locations.

6.1.3. User actions

- Select the parcel/location for the weather forecast.
- Review the weather forecast by day and time.
- Switch to spray forecast when planning crop protection operations.
- Switch to UAV flight forecast when planning drone missions.